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Abstract

We introduce the Recursive Portfolio Machine (RPM), an asset-pricing framework for learning the characteristic directions most relevant for the stochastic discount factor (SDF). Our theory shows that the risk-weighted gradient outer product (RWGOP) recovers the pricing-relevant subspace of characteristic space, while its eigenvectors define principal pricing directions that rank this subspace by pricing importance. The RPM operationalizes this theory by first estimating the RWGOP and then using it within a closed-form large factor model to construct the SDF. In U.S. equity data, the RPM improves out-of-sample performance relative to benchmark models. Low-rank versions retain much of the full-rank performance, suggesting that the pricing-relevant space is effectively low-dimensional.

Keywords: asset pricing; stochastic discount factor; firm characteristics; feature learning; principal pricing directions; kernel methods

1 Introduction

Empirical asset pricing faces a central problem: determining which directions in firm characteristics drive expected returns. Classical factor models address this problem by fixing a small set of prespecified characteristics and estimating linear portfolio weights on those characteristics. Because these weights are transparent, the models provide a clear interpretation of which characteristic directions matter for pricing. Their low-dimensional structure, however, often limits out-of-sample performance. Recent machine-learning methods address the same problem in high-dimensional characteristic spaces and often achieve stronger out-of-sample results because they allow for rich nonlinear interactions. Yet these methods are difficult to interpret, since they learn characteristic transformations and portfolio weights jointly through complex optimization.

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This paper argues that the central problem is to identify a portfolio rule and the pricing-relevant subspace on which it operates. That is, if the SDF is represented by a tradable portfolio with characteristic-based weights, the empirical task is to determine which linear combinations of firm characteristics are sufficient for pricing.

Our theory shows that the risk-weighted gradient outer product (RWGOP) recovers this pricing-relevant subspace. The RWGOP measures how strongly portfolio returns respond to perturbations in characteristic directions after weighting those responses by return risk. We show that the dimension of the subspace on which a portfolio rule lives coincides with the subspace spanned by the RWGOP. The RWGOP, therefore, identifies the characteristic combinations sufficient to represent the portfolio rule.

Within the pricing-relevant subspace, we define principal pricing directions that rank the characteristic combinations relevant for pricing. These directions are the eigenvectors of the RWGOP and provide an ordered basis for the subspace it identifies. This also raises the question of how well the portfolio rule can be approximated using only the leading directions. We show that the resulting approximation error is governed by the residual eigenvalue mass outside the retained directions. When the spectrum is concentrated, the pricing-relevant subspace is effectively low-dimensional, and the leading directions retain most of the pricing-relevant variation.

We use this insight to construct the Recursive Portfolio Machine (RPM). The RPM learns the pricing-relevant geometry of characteristic space from the sample RWGOP and then uses that geometry to construct the SDF. To implement this idea, we build on the closed-form large factor model of Kelly et al. (2026), which represents the SDF through a portfolio kernel that measures similarity across market states. Our contribution is to make that kernel adaptive to the pricing problem. That is, rather than imposing a fixed metric on characteristic space, the RPM uses the sample RWGOP to learn which characteristic directions are most relevant for pricing and then reweights the kernel toward those directions.

The recursive structure of the RPM formalizes a procedure that researchers often follow informally. In practice, researchers rarely fix a characteristic representation once and for all. Instead, they begin with a broad set of characteristics, study which combinations appear most relevant for pricing, and then revise the representation accordingly. The RPM follows the same logic. It uses the RWGOP to identify the pricing-relevant subspace, updates how similarity is measured across characteristics, and then re-estimates the SDF under the new geometry. It then repeats this process, so characteristic discovery and portfolio construction evolve jointly through recursion.

This recursive approach is closely related to recent work on feature learning in machine learning. Radhakrishnan et al. (2022) show that useful feature representations can be learned through recursive updates based on gradient outer products. The RPM adapts this insight to asset pricing. In our setting, the recursion is governed by the SDF objective, so the model learns the geometry of characteristic space from return-weighted gradients of the portfolio rule.

Our empirical results show that learning the characteristic geometry from the RWGOP improves out-of-sample SDF performance. Using U.S. equity data and a large panel of firm characteristics, we find that the RPM outperforms large-factor benchmarks based on fixed random features over a broad range of prediction regularization values. These gains persist when the regularization parameter is selected in real time using only information available at the time of investment. This evidence shows that successful SDF estimation depends on identifying the pricing-relevant subspace and that the RWGOP helps recover that subspace from the data.

Our empirical results also support the theory of principal pricing directions. We find that the learned RWGOP is effectively low-dimensional, in the sense that low-rank versions of the model retain much of the performance of the full-rank specification. This pattern implies that the pricing-relevant variation captured by the RWGOP is concentrated in its leading directions. The evidence, therefore, suggests that a relatively small number of learned characteristic combinations span most of the pricing-relevant variation in the portfolio rule.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 sets up the economic environment, introduces the large factor model, and motivates the RPM. Section 4 develops the theory of the RWGOP and principal pricing directions, and constructs the RPM. Section 5 evaluates the model using U.S. equity data. Section 6 concludes.

2 Related Literature

Factor models and the cross-section of returns. Empirical asset pricing has long organized the search for the stochastic discount factor (SDF) around parsimonious factor structures (Ross, 2013; Fama and French, 1992, 1993, 2015; Kozak et al., 2018, 2020a). A growing body of evidence, however, suggests that a small set of characteristics does not fully capture pricing variation. Harvey et al. (2016) document over 300 candidate factors in the published literature, McLean and Pontiff (2016) show that return predictors decay significantly after publication, and Hou et al. (2020) find that many anomalies fail to replicate under stricter controls. These findings suggest that the effective dimension of the pricing kernel is large and that no fixed low-dimensional factor set reliably spans the SDF across samples and time periods.

Large factor models and the virtue of complexity. This evidence has shifted attention toward high-dimensional factor models that combine very large collections of characteristic-managed portfolios. Kozak et al. (2020b) and Kelly et al. (2023b) show that ridge-regularized combinations of many characteristics outperform sparse alternatives, while Lettau and Pelger (2020) show that pricing information is dispersed across many weak directions rather than concentrated in a few strong factors. Didisheim et al. (2024); Hellum et al. (2026) establish a virtue-of-complexity result. In the limit of infinitely many nonlinear features, the resulting large factor portfolio converges to the conditional SDF, and out-of-sample Sharpe ratios increase with the number of factors even when it greatly exceeds the sample size. Kelly et al. (2024); Kelly and Malamud (2025) provide theoretical foundations for this result by showing that ridge regression in high dimensions achieves risk reduction through endogenous shrinkage governed by the spectrum of the factor covariance matrix. Kelly and Malamud (2025), Chernov et al. (2025), and Kelly et al. (2026) further characterize how spectral complexity and alignment determine achievable out-of-sample performance. The RPM operates within this large-factor framework, but differs from models based on fixed or random transformations because it learns the pricing-relevant geometry of characteristic space and uses that geometry to construct the SDF.

Machine learning in asset pricing. A parallel literature applies neural networks and other flexible machine-learning methods to return prediction and SDF estimation. Kelly et al. (2020) document large out-of-sample gains from tree-based and neural-network methods relative to linear benchmarks. Gu et al. (2020) compare a broad set of machine-learning methods applied to firm characteristics and find that deep networks deliver the strongest and most consistent performance. Chen et al. (2023) embed a deep network within a no-arbitrage pricing framework, and Fan et al. (2022) impose structural cross-equation restrictions to sharpen neural-network-based SDF estimates. Bryzgalova et al. (2025), Bryzgalova et al. (2023), and Cong et al. (2025) study tree-based approaches to factor selection, while Kelly et al. (2023a) connect deep learning to latent factor structures. A common limitation of fully trained deep networks is that their economic content is difficult to characterize analytically, because characteristic transformations and portfolio rules are learned jointly within a blackbox optimization. The RPM addresses this problem by separating the learning of the pricing-relevant geometry from the construction of the SDF while maintaining a closed-form representation at every step.

Kernel methods and the portfolio kernel. Kernel methods provide a tractable way to construct large collections of nonlinear characteristic-managed portfolios while retaining closed-form solutions. Kozak (2020) show that kernel ridge regression delivers strong cross-sectional return predictions when applied to rich characteristic sets. Didisheim et al. (2024) analyze the infinite-feature limit and connect it to large factor representations. Kelly et al. (2026) introduce the Portfolio Kernel as a unified object that encodes both the transformation of characteristics and their cross-sectional aggregation into an SDF. They further show that SDFs estimated from trained neural networks admit an exact kernel representation governed by the Portfolio Tangent Kernel. The RPM builds on this closed-form kernel representation and replaces the fixed metric of the baseline model with one learned from the pricing problem.

Feature learning and recursive feature machines. Recent work in machine learning studies how deep neural networks (DNNs) adapt feature representations during training. Yang et al. (2023) show that DNNs learn features by adaptively learning the Neural Tangent Kernel (NTK; see Jacot et al. (2018)). Kelly et al. (2026) extend this idea to asset pricing and show that DNNs learn to price assets through the Portfolio Tangent Kernel (PTK), the finance counterpart of the NTK. The mechanism through which the NTK and PTK are learned became clearer after Radhakrishnan et al. (2022), who show that DNNs learn features by aligning the weight Gram matrix at a given layer with the average outer product of the gradient of the network output with respect to its inputs. They also show that alternating between kernel estimation and gradient-based reweighting yields a Recursive Feature Machine (RFM) that matches or surpasses fully trained networks on tabular benchmarks. The RPM is the asset-pricing analogue of this idea and sheds light on the PTK learning mechanism of Kelly et al. (2026). Rather than optimizing a prediction loss, the recursive update in the RPM is driven by the SDF objective, so the RWGOP increases the weight placed on characteristic directions that matter more for pricing.

3 Framework

3.1 Economic Environment

We study a panel of N_t stocks observed monthly. At date t, stock i has excess return $R_{i,t+1}$ and a d-dimensional vector of firm characteristics $X_{i,t} \in \mathbb{R}^d$. Let

$$R_{t+1} \in \mathbb{R}^{N_t}, \qquad X_t \in \mathbb{R}^{N_t \times d}$$

collect the cross-section of next-period returns and current characteristics.

Our object of interest is the tradable SDF. In this setting, the SDF can be written as

$$M_{t+1} = 1 - \pi_t' R_{t+1}, \tag{1}$$

where $\pi_t \in \mathbb{R}^{N_t}$ denotes the vector of portfolio weights across stocks. Under standard mean–variance preferences, the optimal portfolio is the conditional Markowitz portfolio,

$$\pi_t^* = E_t[R_{t+1} R_{t+1}']^{-1} E_t[R_{t+1}], \tag{2}$$

where $E_t[\cdot]$ denotes conditional expectation given the information available at time t. This portfolio solves

$$\max_{\pi_t} E_t[U(\pi_t' R_{t+1})], \qquad U(x) = x - \tfrac{1}{2}x^2. \tag{3}$$

Thus, estimating the SDF is equivalent to identifying the conditionally efficient portfolio of excess returns.

The challenge is that the conditional moments in (2) are not observed directly. We therefore use firm characteristics as conditioning information and restrict portfolio weights to be functions of the cross-section of characteristics

$$\pi_t = \pi(X_t).$$

This yields the characteristic-based SDF representation

$$M_{t+1} = 1 - \pi(X_t)' R_{t+1}.$$

The central modeling problem is therefore to learn a map from characteristics to tradable portfolio weights.

Using the law of iterated expectations, the conditional portfolio-choice problem in (3) is equivalent to the unconditional mean-squared-error problem

$$\min_{\pi(\cdot)} E\left[(1 - \pi(X_t)' R_{t+1})^2\right]. \tag{4}$$

Equation (4) shows that the goal is to find the characteristic-based portfolio rule that best approximates the tradable SDF.

To operationalize (4), we approximate the portfolio rule $\pi(X_t)$ with a flexible function class $f(X_t; \theta)$ and use the penalized sample analogue

$$\min_{\theta} \frac{1}{T} \sum_{t=1}^{T} \big(1 - f(X_t; \theta)' R_{t+1}\big)^2 + z\|\theta\|^2, \tag{5}$$

as our objective function for learning the optimal characteristic-based portfolio rule. Here $z > 0$ is a ridge penalty that regularizes the estimation problem and stabilizes the fitted portfolio rule when the function class is high-dimensional or highly flexible.

This formulation separates the SDF problem into two distinct tasks. First, we must choose a function class $f(\cdot; \theta)$ rich enough to generate economically meaningful managed portfolios from characteristics. Second, we must determine which directions in characteristic space are most relevant for pricing, so that the model emphasizes informative characteristic combinations and shrinks uninformative ones. The first task motivates the large-factor and kernel representation introduced next. The second task motivates the Recursive Portfolio Machine developed in Section 4.

3.2 Large Factor Models

The objective in (5) requires a function class that maps firm characteristics into portfolio weights in a flexible but tractable way. We do this by constructing the portfolio rule from a large collection of factors. This keeps the SDF in the form of a portfolio return while allowing the portfolio rule to depend nonlinearly on characteristics.

Fix a nonlinear activation function $\phi : \mathbb{R} \to \mathbb{R}$ and draw random feature vectors $W_k \in \mathbb{R}^d$, for $k = 1, \ldots, P$, independently from some distribution, where P denotes the number of factors. For each draw W_k, define the factor return

$$F_{k,t+1} = \sum_{i=1}^{N_t} \phi(X_{i,t}' W_k)\, R_{i,t+1}. \tag{6}$$

The kth factor is the return on a long–short portfolio whose weights are determined by the nonlinear signal $\phi(X_{i,t}' W_k)$.

We then approximate the portfolio rule $\pi(X_t)$ by a linear combination of these factor weights,

$$f(X_t;\theta) = \sum_{k=1}^{P} \theta_k \phi(X_t W_k).$$

Under this specification, the optimal coefficient vector that solves the sample objective in (5) is

$$\theta(z;P) = \left(zI + T^{-1} \sum_{t=1}^{T} F_t F_t' \right)^{-1} T^{-1} \sum_{t=1}^{T} F_t, \tag{7}$$

and the associated SDF return in the random features model is

$$R_{t+1}^{\text{RF}} = \theta(z;P)' F_{t+1}. \tag{8}$$

This finite-P formulation makes clear how model complexity enters the problem. As P increases, the SDF is allowed to combine a richer collection of factors. Recent work shows that out-of-sample performance can continue to improve as the number of factors becomes very large (Didisheim et al., 2024). This motivates studying the limit as $P \to \infty$.

To characterize the $P \to \infty$ limit of the random features model, we first define the Portfolio Kernel introduced by Kelly et al. (2026).

Definition 1 (Portfolio Kernel, Kelly et al. 2026) *Let*

$$Y_t = (R_t, X_{t-1}) \in \mathbb{R}^{N_{t-1}(d+1)} \tag{9}$$

denote the market state at time t. Given a kernel K and two market states $Y = (R, X)$ and $\tilde{Y} = (\tilde{R}, \tilde{X})$, define the cross-stock kernel matrix

$$K(X, \tilde{X}) = (K(X_i, \tilde{X}_j)) \in \mathbb{R}^{N \times \tilde{N}}.$$

The corresponding Portfolio Kernel is

$$\mathbb{K}(Y; \tilde{Y}) = R' K(X, \tilde{X}) \tilde{R}. \tag{10}$$

The Portfolio Kernel is the market-state analogue of the stock-level kernel K. The kernel $K(x, \tilde{x})$ measures similarity between two characteristic vectors, while $\mathbb{K}(Y; \tilde{Y})$ aggregates these stock-level similarities across all stocks in the two market states using returns. As a result, $\mathbb{K}(Y;\tilde{Y})$ is a scalar return that summarizes how strongly the two market states are connected through characteristic similarity.

The following theorem, proved in Kelly et al. (2026), characterizes the $P \to \infty$ limit of the random features model.

Theorem 1 (Large-Factor-Model SDF, Kelly et al. 2026) *As $P \to \infty$, the random features model SDF return in (8) converges to a large factor model with a closed-form representation in terms of the Portfolio Kernel. Let*

$$Y_{IS} = (Y_t)_{t=1}^{T}$$

denote the collection of in-sample market states, and define the $T \times T$ in-sample Portfolio Kernel matrix by

$$\mathbb{K}_{IS} = \big(\mathbb{K}(Y_{t_1};Y_{t_2})\big)_{t_1,t_2=1}^{T}. \tag{11}$$

Then

$$R_{T+1}^{\mathrm{LFM}}(z) = \mathbb{K}(Y_{T+1};Y_{IS})\,\xi = \sum_{t=1}^{T} \mathbb{K}(Y_{T+1};Y_t)\,\xi_t, \tag{12}$$

where

$$\xi(T,z) = \frac{1}{T}\left(zI + T^{-1}\mathbb{K}_{IS}\right)^{-1}\mathbf{1}. \tag{13}$$

Equivalently, the portfolio rule can be written as

$$f_T(x) = \sum_{t=1}^{T} K(x, X_{t-1})\, R_t\, \xi_t(T,z). \tag{14}$$

Theorem 1 is the large factor model representation that we use throughout the paper. The stock-level kernel $K(x,\tilde{x})$ determines how similarity between firm characteristics is measured, while the Portfolio Kernel $\mathbb{K}(Y;\tilde{Y})$ aggregates those similarities into returns at the level of market states. The coefficients ξ_t determine how past market states enter the final SDF return.

We now specialize the stock-level kernel K to the ReLU random features construction, following Didisheim et al. (2024). In the limit as $P \to \infty$, the corresponding kernel is the arc-cosine kernel.

Theorem 2 (Arc-cosine kernel) *Let* $W_k \sim \mathcal{N}(0, I_d)$ *independently and define*

$$K_P(x, y) = \frac{1}{P} \sum_{k=1}^{P} \phi(W_k' x) \phi(W_k' y).$$

As $P \to \infty$*,*

$$K(x, y) = \frac{\|x\| \, \|y\|}{2\pi} \Big(\sin\vartheta + (\pi - \vartheta) \cos\vartheta \Big), \qquad \vartheta = \arccos\left(\frac{x'y}{\|x\| \, \|y\|} \right). \tag{15}$$

The arc-cosine kernel in (15) is the baseline kernel that we use in the remainder of the paper.

3.3 Motivating the Recursive Portfolio Machine

The baseline arc-cosine kernel in (15) fixes the geometry of characteristic space in advance. Similarity between two firms is determined by the Euclidean inner product $x'y$, so each characteristic direction enters symmetrically. For the pricing problem, however, there is no reason to expect all characteristic directions to be equally relevant for the SDF.

This suggests a more flexible kernel. Instead of fixing the geometry of characteristic space ex ante, we allow the model to reweight characteristic directions according to their pricing relevance. We do this by replacing the Euclidean inner product $x'y$ with a quadratic form defined by a symmetric positive semidefinite matrix $M \in \mathbb{R}^{d \times d}$. This gives the kernel

$$K_M(x, y) = \frac{\sqrt{x'Mx} \, \sqrt{y'My}}{2\pi} \Big(\sin\vartheta_M + (\pi - \vartheta_M) \cos\vartheta_M \Big), \tag{16}$$

where

$$\vartheta_M = \arccos\left(\frac{x'My}{\sqrt{x'Mx} \, \sqrt{y'My}} \right).$$

When $M = I_d$, (16) reduces to the arc-cosine kernel in (15).

The matrix M determines how similarity is measured across characteristic directions. Directions associated with larger eigenvalues of M have greater influence on the kernel, while directions associated with smaller eigenvalues have less influence. Equivalently, M determines which linear combinations of characteristics receive more weight in portfolio formation.

This is the basic idea behind the Recursive Portfolio Machine. Rather than fixing M in advance, we let the data learn it. In the next section, we introduce an iterative update for M that increases weight on characteristic directions that contribute more to the SDF return.

4 The Recursive Portfolio Machine

4.1 Problem Setup

Throughout this section, let $\mathcal{S} \subset \mathbb{R}^d$ denote the characteristic space, assumed to be open and connected. For each stock i, let $x_i \in \mathcal{S}$ denote its characteristic vector, and let

$$X = (x_i)_{i=1}^N \in \mathcal{S}^{\otimes N}$$

denote the full cross-section of characteristics. Let $R \in \mathbb{R}^N$ denote the corresponding vector of returns, and let $f : \mathcal{S} \to \mathbb{R}$ denote the portfolio rule that maps a stock's characteristics into its portfolio weight. We write

$$f(X) = (f(x_i))_{i=1}^N$$

for the vector of portfolio weights across the cross-section, and

$$Jf(X) \ := \ \left(\frac{\partial f(x_i)}{\partial x_i} \right)_{i=1}^N \in \mathbb{R}^{N \times d}$$

for the Jacobian of the portfolio rule with respect to characteristics. We also define the conditional second-moment matrix

$$\Sigma(X) \ := \ E[RR' \mid X] \in \mathbb{R}^{N \times N}.$$

The previous subsection argued that the geometry of characteristic space should be learned from the pricing problem rather than fixed in advance. This raises the central question of the Recursive Portfolio Machine: which directions in characteristic space are actually relevant for pricing? Although the portfolio rule f may in principle depend on the full d-dimensional characteristic vector x, it is natural to expect that pricing-relevant variation is concentrated in a much smaller set of linear combinations of characteristics. If so, then the portfolio rule depends only on a low-dimensional linear subspace of characteristic space. We formalize this idea as follows.

Definition 2 (Low-Rank Portfolio Rule) *We say that f has rank r on $\mathcal{S}$ if there exists* *a matrix* $B \in \mathbb{R}^{d\times r}$ *with* $r < d$ *and a function* $g : B'\mathcal{S} \to \mathbb{R}$ *such that*

$$f(x) = g(B'x) \quad \text{for all } x \in \mathcal{S}.$$

A rank-r portfolio rule depends on the full characteristic vector x only through the r linear combinations in $B'x$. Equivalently, f depends on x only through its projection onto the r-dimensional subspace spanned by the columns of B. Once that subspace is known, all remaining directions in characteristic space are irrelevant for the portfolio rule.

The problem we therefore aim to solve is to recover the minimal low-dimensional subspace of characteristic space that is sufficient to represent the portfolio rule.

4.2 Risk-Weighted Gradient Outer Product

We now introduce the population object that solves the low-dimensional subspace problem.

Definition 3 (Risk-Weighted Gradient Outer Product) *Assuming* $E\|Jf(X)\|_F^2 < \infty$*, the risk-weighted gradient outer product (RWGOP) is*

$$M_* \;:=\; E[Jf(X)'\,RR'\,Jf(X)] \in \mathbb{R}^{d\times d}. \tag{17}$$

The RWGOP measures how strongly portfolio returns respond to characteristic directions after weighting those responses by return risk. The Jacobian $Jf(X)$ captures the local sensitivity of portfolio weights to perturbations in characteristics across stocks. In fact, M_* is the covariance matrix of portfolio returns $Jf(X)'R$. As a result, M_* places large weight on directions in characteristic space that both change portfolio weights and induce substantial variation in portfolio returns. Directions that have little effect on return variation receive little weight. The RWGOP, therefore, summarizes which linear combinations of characteristics are most relevant for pricing.

The next lemma shows that the RWGOP recovers the pricing-relevant subspace itself.

Lemma 1 (RWGOP Characterization of the Pricing Subspace) *Let* (R, X) *be a random vector satisfying the setup in Section 4.1. Assume that* X *admits a density on* $\mathcal{S}^{\otimes N}$ *which is positive Lebesgue-a.e., that* f *is real analytic, and that*

$$\Sigma(X) \;\succeq\; \kappa I_N \qquad \text{almost surely} \tag{18}$$

for some $\kappa > 0$.

Let $\operatorname{rank}(M_*) = r$ *and let* $M_* = UDU'$ *be an eigendecomposition with* $U \in \mathbb{R}^{d\times r}$ *having orthonormal columns spanning* $\operatorname{Im}(M_*)$*, and let* $P := UU'$ *be the orthogonal projector onto* $\operatorname{Im}(M_*)$*.*

(1) (Existence of an analytic reduction.) There exists a real analytic map $g : U'\mathcal{S} \to \mathbb{R}$ *such that*

$$f(x) \ = \ g(U'x) \qquad \text{for all } x \in \mathcal{S}.$$

In particular, portfolio returns satisfy $R'f(X) = R'g(U'X)$ *almost surely.*

(2) (Converse / rank bound.) Conversely, if there exist $r \le d$*, a matrix* $B \in \mathbb{R}^{d\times r}$ *with orthonormal columns, and a (measurable) map* $g : B'\mathcal{S} \to \mathbb{R}$ *such that*

$$f(x) = g(B'x) \quad \text{for all } x \in \mathcal{S},$$

then $\operatorname{rank}(M_*) \le r$*.*

(3) (Full rank under non-degeneracy of reduced gradients.) Assume in addition that g *is real analytic and that for every nonzero* $a \in \mathbb{R}^r$*,*

$$\Pr\big(a'Jg(B'X)'\Sigma(X)Jg(B'X)a > 0\big) > 0.$$

Then $\operatorname{rank}(M_*) = r$*.*

Lemma 1 makes the solution precise. If the portfolio rule depends only on a subspace of dimension r, then part (2) shows that the RWGOP has rank at most r. If the RWGOP has rank r, then part (1) shows that the portfolio rule f can be written as a function of $U'x$, where the columns of U span $\operatorname{Im}(M_*)$. Under the nondegeneracy condition in part (3), these two notions coincide exactly, so the rank of M_* equals the dimension of the minimal pricing-relevant subspace. Estimating M_* therefore recovers both the dimension of that subspace and the linear combinations of characteristics that span it. Once we know M_*, we know the low-dimensional subspace on which f depends and can represent f through that subspace.

4.3 Principal Pricing Directions

The RWGOP recovers the pricing-relevant low-dimensional subspace. The next step is to understand how that subspace is organized internally. In particular, once the relevant subspace has been identified, we would like to know which directions within it matter most for pricing and how much is lost by retaining only the leading ones.

Definition 4 (Principal Pricing Directions) *The eigenvectors of M_* are the principal pricing directions of the portfolio function f.*

The principal pricing directions provide an ordered basis for the pricing-relevant low-dimensional subspace. They are the principal components of the managed portfolios $Jf(X)'R$. The leading eigenvector identifies the direction that carries the most pricing-relevant variation, and the remaining eigenvectors identify orthogonal directions that carry progressively less. They therefore describe how pricing information is organized within the subspace recovered by the RWGOP.

This ordering immediately raises the relevant approximation question. If we retain only the top p principal pricing directions, how much pricing-relevant variation do we lose? The answer is given by the eigenvalue spectrum of M_*. Let

$$M_* = UDU'$$

denote the spectral decomposition of the RWGOP, with eigenvalues ordered as

$$D = \operatorname{diag}(\lambda_1(M_*), \ldots, \lambda_d(M_*)), \qquad \lambda_1(M_*) \geq \cdots \geq \lambda_d(M_*) \geq 0,$$

and define

$$\Lambda_{-p}(M_*) \; = \; \sum_{k=p+1}^{d} \lambda_k(M_*)$$

to be the residual eigenvalue mass beyond the leading p directions. The following corollary gives the precise approximation bound implied by this residual eigenvalue mass.

Corollary 3 (Approximation by Principal Pricing Directions) *Suppose, in addition to the setup of Section 4.1, that*

$$X = A^{1/2} Y B^{1/2}, \qquad \operatorname{vec}(Y) \sim N(0, I_{Nd}), \tag{19}$$

where $A \in \mathbb{R}^{N \times N}$ and $B \in \mathbb{R}^{d \times d}$ are symmetric positive definite, and that

$$\Sigma \; = \; E[RR'|X] - E[R|X]E[R|X]'$$

is constant. Let $M_* = UDU'$ be the spectral decomposition of the RWGOP, and partition

$U = [U_p \; U_{-p}]$, where U_p contains the leading p eigenvectors. Define

$$f_p(XU_p) := E[f(X) \mid XU_p].$$

Then

$$E[\mathrm{Var}[(f(X) - f_p(XU_p))'R|X]] \leq \lambda_1(A)\lambda_1(B)\Lambda_{-p}(M_*). \tag{20}$$

Corollary 3 gives the eigenvalue spectrum a direct economic interpretation. The approximation error on the left-hand side measures the expected residual variation in portfolio returns when the full portfolio rule f is replaced by its projection onto the span of the leading p principal pricing directions. In other words, it measures how much pricing-relevant variation is lost when we discard the trailing directions in characteristic space. The corollary shows that this loss is controlled by the residual eigenvalue mass $\Lambda_{-p}(M_*)$. When the trailing eigenvalues are small, the omitted directions contribute little to portfolio-return variation, so the portfolio rule is well approximated by the leading principal pricing directions alone.

4.4 The Recursive Portfolio Machine

The previous subsections characterize the object that the model should learn. The RWGOP recovers the pricing-relevant low-dimensional subspace, and its eigenvectors identify the principal pricing directions within that subspace. We now use this theory to construct the Recursive Portfolio Machine (RPM), which learns the relevant geometry of characteristic space and uses it to form the SDF.

The theory suggests that the portfolio rule should depend on characteristics through a linear transformation that reflects the pricing-relevant low-dimensional subspace. If M_* denotes the population RWGOP, then a natural class of portfolio rules is

$$f_{M_*}(x; \theta) = f(M_*^{1/2} x; \theta), \tag{21}$$

where $f(\cdot; \theta)$ is a flexible function class. In this specification, the matrix M_* determines which linear combinations of characteristics are emphasized, while the function class $f(\cdot; \theta)$ maps the transformed characteristics into portfolio weights. In practice, however, the population object M_* is unknown, so both the geometry of characteristic space and the portfolio rule must be estimated from the data.

In the RPM, the function class we consider is the large-factor model of Theorem 1. This choice fits the general form in (21) exactly. For the arc-cosine kernel, the reweighted kernel satisfies

$$K_M(x, y) = K(M^{1/2}x, M^{1/2}y),$$

so applying the kernel under metric M is equivalent to applying the baseline kernel to linearly transformed characteristics. Under this specification, the portfolio rule takes the form

$$f_T(x) = \sum_{t=1}^{T} K_M(x, X_{t-1})\, R_t\, \xi_t,$$

so the matrix M determines how similarity is measured across characteristic directions. This large-factor specification is especially convenient in our setting because the SDF remains available in closed form, and the gradient of the portfolio rule with respect to characteristics can be computed analytically.

The key difficulty is that M and the portfolio rule are interdependent. The RWGOP (17) is defined in terms of the portfolio rule, but the portfolio rule itself depends on the metric M. We therefore estimate them jointly through a recursive procedure. Given a fitted portfolio rule f_T, we update the metric according to the finite-sample analog of (17):

$$M \leftarrow \frac{1}{T} \sum_{t=1}^{T} Jf_T(X_{t-1})' R_t R_t' Jf_T(X_{t-1})\,. \tag{22}$$

This recursive procedure is motivated by recent work in machine learning that links feature learning to gradient structure. In particular, Radhakrishnan et al. (2022) show that useful feature representations can be learned by recursively updating a matrix built from gradient outer products. The RPM adapts this idea to asset pricing. In our setting, the recursively updated matrix is the RWGOP, and its leading eigenvectors have the economic interpretation of principal pricing directions.

We formalize the resulting RPM in Algorithm 1.

Algorithm 1 Recursive Portfolio Machine (RPM)

```
**Require:** Data $\{(X_t, R_{t+1})\}_{t=1}^{T}$, ridge penalty $z$, iterations $G$
Initialize $M_0 \leftarrow I_d$
for $g = 1, \dots, G$ do
    Compute $\mathbb{K}_{IS}$ under metric $M_{g-1}$
    $\xi \leftarrow \frac{1}{T}\left(zI + T^{-1}\mathbb{K}_{IS}\right)^{-1}\mathbf{1}$
    $f_T(x) = \sum_{t=1}^{T} K_{M_{g-1}}(x, X_{t-1}) R_t \xi_t$
    $M_g \leftarrow \frac{1}{T}\sum_{\tau=1}^{T} \left(Jf_T(X_{\tau-1})' R_\tau\right)\left(Jf_T(X_{\tau-1})' R_\tau\right)'$
end for
return $M_G$
```

The RPM therefore operationalizes the theory developed above. It estimates the pricing-relevant low-dimensional subspace through the recursive RWGOP update and then uses the associated reweighted kernel to construct the SDF. The converged matrix M_I determines which characteristic directions receive the greatest emphasis in portfolio formation, while its eigenvalue spectrum measures how concentrated pricing information is within that low-dimensional subspace.

5 Empirics

5.1 Data

We use a comprehensive dataset of U.S. monthly equity characteristics and returns from Jensen et al. (2023), available via WRDS. The dataset spans 1963 to 2024 and consists of 153 characteristics that have been shown to predict stock returns in the literature.

To obtain a stock–month panel with a stable set of characteristics across time and firms, we adopt the filtering approach proposed in Didisheim et al. (2024). First, we restrict the sample to ordinary common shares (CRSP share codes 10, 11, or 12) listed on the NYSE, Nasdaq, or AMEX (CRSP exchange codes 1, 2, or 3). Second, we retain only characteristics with sufficient data availability, excluding variables with more than 30% missing observations over the full sample; this leaves 132 characteristics out of an initial set of 153. Third, to limit the influence of extremely small and illiquid firms, we remove nano-cap stocks whose market capitalization falls below the 1st percentile of NYSE market capitalizations at each date. Fourth, we discard stock–month observations with missing values for more than one-third of the retained characteristics. Finally, the remaining characteristics are rank-normalized cross-sectionally each month by rescaling ranks to the interval $[-0.5, 0.5]$ and imputing any remaining missing values with the cross-sectional median (zero by construction). After preprocessing, the characteristics of the N_t stocks at date t form an $N_t \times 132$ matrix X_t, which serves as the conditioning information set, and we denote by $R_{t+1} = (R_{i,t+1})_{i=1}^{N_t}$ the vector of next-month excess returns.

To abstract from liquidity differences while isolating the role of statistical complexity as proposed in Kelly et al. (2026), we conduct the analysis separately across market capitalization groups. Following Jensen et al. (2023), stocks are sorted each period using NYSE breakpoints into mega-cap (top 20%), large-cap (80th–50th percentiles), small-cap (50th–20th percentiles), and micro-cap (20th–1st percentiles) portfolios. We also analyze the full universe of firms above the 1% NYSE size cutoff, which we refer to as “All”.

5.2 Benchmark Models

We compare RPM to two benchmark SDF models estimated under the same rolling-window out-of-sample protocol. In all benchmark forecasts, the prediction window length is selected from

$$T \in \{12, 60, 120, 240, 360\}, \tag{23}$$

and the final prediction shrinkage parameter is selected from

$$z \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}\}. \tag{24}$$

We report results for the target universes

$$\{\text{All}, \text{Mega}, \text{Large}, \text{Small}, \text{Micro}\}.$$

For each benchmark, factor construction and rolling prediction are both carried out within the target universe.

The linear-ridge benchmark, denoted LR, constructs characteristic-managed factor returns from the raw linear characteristics (Brandt et al., 2009). Its managed factor returns are

$$F_{t+1}^{\text{LR}} = N_t^{-1/2} R_{t+1}' X_t, \tag{25}$$

and its rolling-window Markowitz-efficient SDF payoff is

$$R_{t+1}^{\text{LR}}(T, z) = \theta_t(T, z)' F_{t+1}^{\text{LR}}, \tag{26}$$

where $\theta_t(T, z)$ solves the ridge-regularized Markowitz problem over the trailing T months as in (7).

The random-feature benchmark, denoted RF, constructs nonlinear factor returns from characteristics using a high-dimensional ReLU random-feature expansion as in Didisheim et al. (2024):

$$S_t = \phi(X_t) \in \mathbb{R}^{N_t \times P}, \qquad \phi(x) = \mathrm{ReLU}(x\Gamma), \tag{27}$$

where $\Gamma \in \mathbb{R}^{d \times P}$ has i.i.d. standard normal entries. The associated managed factor returns are

$$F_{t+1}^{\mathrm{RF}} = N_t^{-1/2} R'_{t+1} S_t, \tag{28}$$

and the corresponding benchmark payoff is

$$R_{t+1}^{\mathrm{RF}}(T, z) = \theta_t(T, z)' F_{t+1}^{\mathrm{RF}}. \tag{29}$$

By Theorem 2, as $P \to \infty$, RF converges to the corresponding arc-cosine kernel large-factor model with $M_0 = I_d$. In this baseline specification, the characteristic metric is the identity matrix, so the kernel does not reweight characteristic space.

All characteristic-managed portfolios are normalized by $N_t^{-1/2}$ so that SDF payoffs are comparable across models and over time. This normalization removes mechanical variation in scale induced by changes in cross-sectional sample size.

We also scale the shrinkage parameter z by the average eigenvalue of the matrix being regularized so that the same grid of values is comparable across models. In primal formulations, this scaling is $\mathrm{tr}(A)/p$ for $A \in \mathbb{R}^{p \times p}$; in dual kernel formulations, it is $\mathrm{tr}(K)/T^2$ for $K \in \mathbb{R}^{T \times T}$. This effective-shrinkage normalization is used throughout the benchmark and RPM forecasts.

5.3 Training the RPM

RPM is trained in two stages. In the first stage, we learn a time-varying RWGOP from rolling monthly out-of-sample gradient updates computed under the RF large-factor model. In the second stage, we use this learned RWGOP to refit the large-factor model in each target evaluation universe. This two-stage design follows Kelly et al. (2026) and separates feature learning from SDF aggregation: the first stage learns the pricing-relevant geometry of characteristic space, and the second stage uses that geometry to construct the final SDF.

In Stage 1, we estimate the RWGOP once using the full cross section of stocks, that is, the “All” universe. This yields a sequence of learned matrices

$$M_{1,t}(z_M, T_u, T_M).$$

In Stage 2, we hold these learned RWGOPs fixed and use them to construct forecasts separately in each target universe

$$\{\mathrm{All}, \mathrm{Mega}, \mathrm{Large}, \mathrm{Small}, \mathrm{Micro}\}.$$

Stage 1: Learning the RWGOP. Stage 1 learns the RWGOP in two steps. First, we fit RF large-factor models with different values of (z_M, T_u) and use them to compute monthly gradient updates. Second, we use these monthly gradient updates to construct learned RWGOPs using different values of T_M.

Computing monthly gradient updates. Fix (z_M, T_u), where z_M is the regularization parameter used to fit the RF large-factor model and T_u is the length of the rolling window used to fit that model. Thus, z_M and T_u play the same roles here that z and T play in the RF benchmark fit in (7). For each month s, define the trailing sample

$$\mathcal{T}_s = \{s - T_u, \dots, s - 1\}.$$

Using only the months in $\mathcal{T}_s$, we fit the RF large-factor model and obtain the fitted portfolio rule

$$f_s(x; T_u, z_M) = \sum_{t \in \mathcal{T}_s} N_t^{-1/2} K(x, X_t) R_{t+1} \xi_t(T_u, z_M). \tag{30}$$

We then evaluate this fitted rule on the next month through its gradient and form the return-weighted monthly gradient vector

$$g_s(T_u, z_M) = \sum_{i=1}^{N_s} R_{i,s+1} \, \nabla f_s(x_{i,s}; T_u, z_M).$$

We treat this return-weighted monthly gradient vector as the RWGOP update for month s, and normalize it by its Euclidean norm,

$$\tilde{g}_s = \frac{g_s}{\|g_s\|_2}.$$

Constructing the learned RWGOP. Fix T_M, the number of monthly RWGOP updates used to construct the learned matrix. For each month t, we define the learned RWGOP as

$$M_{1,t}(z_M, T_u, T_M) = \frac{1}{T_M} \sum_{\tau=1}^{T_M} \tilde{g}_{t-\tau}(T_u, z_M) \tilde{g}_{t-\tau}(T_u, z_M)'.$$

Thus, the learned RWGOP is the average outer product of the most recent T_M normalized monthly RWGOP updates. This matrix summarizes the characteristic directions that repeatedly generate large return-weighted payoff gradients in recent out-of-sample months.

This construction is also computationally efficient because the monthly RWGOP updates can be reused across adjacent rolling windows. For each fixed pair (T_u, z_M), we compute the sequence $\{g_s\}$ once and store it by calendar month. Moving from month t to $t+1$ therefore requires only dropping the oldest gradient object and adding the newest one.

Stage 2: Forecasting with the RWGOP. Stage 2 uses the learned RWGOP $M_{1,t}(z_M, T_u, T_M)$ to construct RPM forecasts. Given this learned RWGOP, we form the associated reweighted kernel $K_{M_{1,t}}(\cdot, \cdot)$ and then construct forecasts in the same way as in the benchmark models. Specifically, we fix (T, z), where T is the length of the rolling prediction window and z is the regularization parameter, and refit the large-factor model over the trailing T months using the kernel $K_{M_{1,t}}(\cdot, \cdot)$. The resulting RPM portfolio rule is

$$f_t^{\mathrm{RPM},1}(x; z_M, T_u, T_M, T, z) = \sum_{\tau=t-T}^{t-1} N_\tau^{-1/2} K_{M_{1,t}}(x, X_\tau)\, R_{\tau+1}\, \xi_{t,\tau}(T, z; K_{M_{1,t}}), \tag{31}$$

and the associated out-of-sample return is

$$R_{t+1}^{\mathrm{RPM},1}(z_M, T_u, T_M, T, z) = N_t^{-1/2} R'_{t+1}\, f_t^{\mathrm{RPM},1}(X_t; z_M, T_u, T_M, T, z). \tag{32}$$

We evaluate Stage 2 over the grid[1]

$$z_M \in \{\pm 10^{-1}, \pm 10^{-2}, \pm 10^{-3}, \pm 10^{-4}\}, \quad T_u \in \{12, 60, 120, 240\}, \quad T_M \in \{12, 60\},$$

together with the benchmark prediction grid for (T, z) in (23)–(24).

---

[1] We allow z_M to take mildly negative values when constructing the RWGOP in Stage 1. Negative values of z_M offset excessive shrinkage in the RF large-factor model used to compute the monthly gradient updates. Recent theory and evidence show that mildly negative regularization can improve out-of-sample SDF performance in overparameterized large-factor models by partially reversing self-induced regularization (Xiong, 2026). Although z_M enters the RWGOP construction rather than the final forecasting step, the same logic motivates including mildly negative values in the Stage 1 grid.

5.4 Performance of the RPM

This subsection evaluates whether the learned RWGOP improves out-of-sample performance relative to the benchmark models. We first study performance as a function of the final prediction regularization parameter z. We then study performance when z is selected in real time using only information available at the time of investment.

All reported strategies are inverse-volatility-weighted ensembles based on a 12-month lagged rolling volatility estimate. For any family of candidate return series $\{R^m_{t+1}\}_{m\in\mathcal{M}}$, we set

$$\omega^m_t = \frac{(\hat{\sigma}^m_t)^{-1}}{\sum_{m'\in\mathcal{M}}(\hat{\sigma}^{m'}_t)^{-1}}, \tag{33}$$

where $\hat{\sigma}^m_t$ is the trailing 12-month realized volatility of candidate m, computed using information available at time t. During the first 12 out-of-sample months, when insufficient volatility history is available, we impose equal weights.

Performance as a function of z. We begin by evaluating performance at fixed values of the final prediction regularization parameter z. For each fixed z, we aggregate across all remaining tuning dimensions while keeping z fixed. This yields a family of implementable out-of-sample strategies indexed by z and isolates the role of final prediction regularization from the other dimensions of the procedure.

At fixed z, the RF benchmark is the inverse-volatility-weighted ensemble of the corresponding rolling-window forecasts:

$$\overline{R}^{\mathrm{RF}}_{t+1}(z) = \sum_T \omega^{\mathrm{RF}}_{t,T}(z)\, R^{\mathrm{RF}}_{t+1}(T, z). \tag{34}$$

At fixed z, our reported RPM strategy combines the RF benchmark with our learned RPM component:

$$\overline{R}^{\mathrm{RPM}}_{t+1}(z) = \omega^{\mathrm{RPM},0}_t(z)\, \overline{R}^{\mathrm{RF}}_{t+1}(z) + \omega^{\mathrm{RPM},1}_t(z)\, \overline{R}^{\mathrm{RPM},1}_{t+1}(z). \tag{35}$$

Here, $\overline{R}^{\mathrm{RPM},1}_{t+1}(z)$ is our learned RPM component. It is the inverse-volatility-weighted ensemble of RPM forecasts across all retained Stage 1 and Stage 2 specifications:

$$\overline{R}^{\mathrm{RPM},1}_{t+1}(z) = \sum_{z_M}\sum_{T_u}\sum_{T_M}\sum_{T} \omega^{\mathrm{RPM},1}_{t,z_M,T_u,T_M,T}(z)\, R^{\mathrm{RPM},1}_{t+1}(z_M, T_u, T_M, T, z). \tag{36}$$

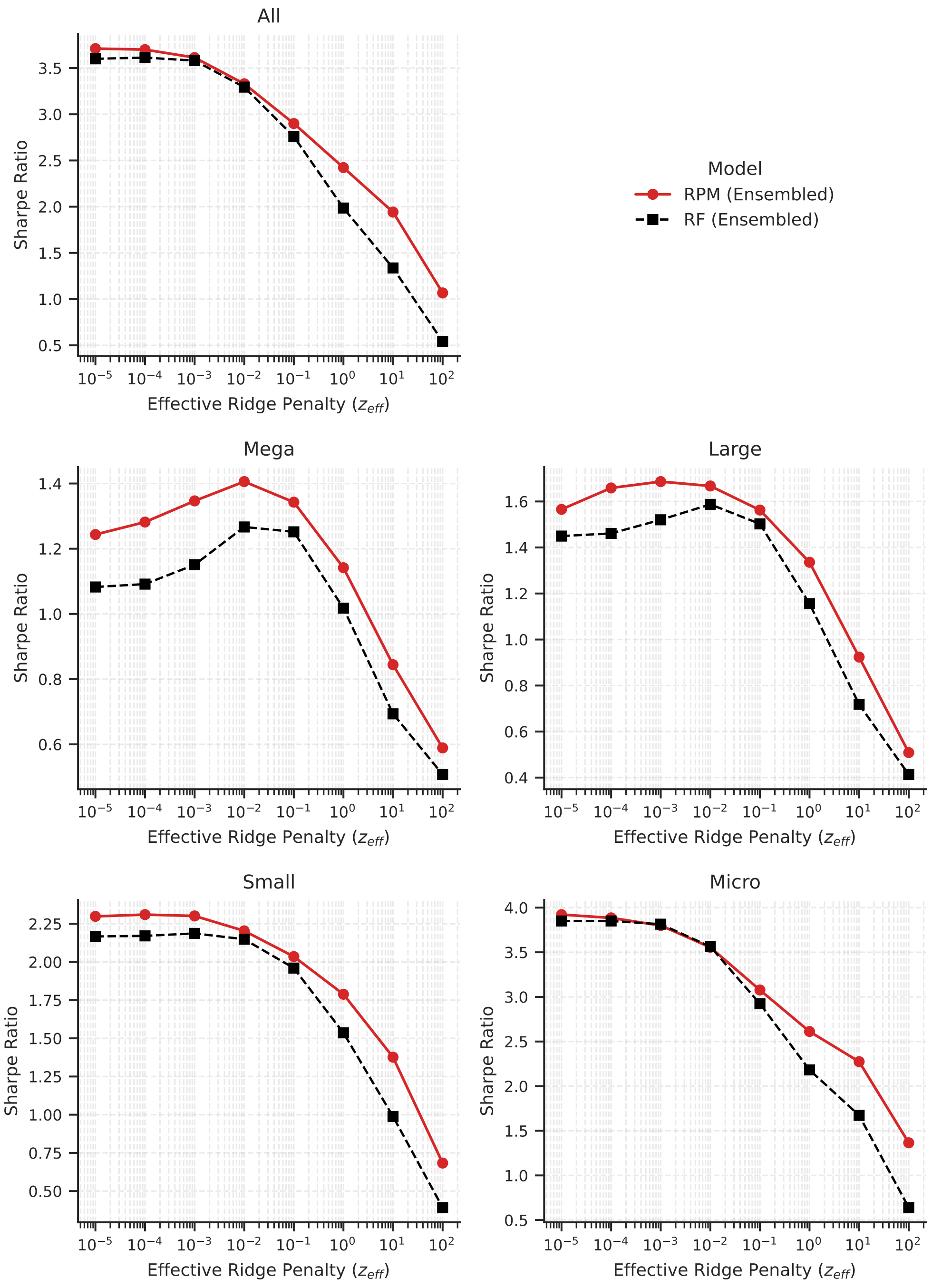

Figure 1: Sharpe Ratio of $\overline{R}^{\text{RPM}}_{t+1}(z)$ as a Function of z

Note: The figure reports the out-of-sample Sharpe ratios of the fixed-z RF and RPM strategies $\overline{R}^{\text{RF}}_{t+1}(z)$ and $\overline{R}^{\text{RPM}}_{t+1}(z)$, defined in (34) and (35), as functions of the prediction regularization parameter z in (24) and separately by evaluation universe. The out-of-sample period is from 1993 to 2024.

Thus, RPM denotes the implementable mixed strategy, while $\overline{R}^{\text{RPM},1}_{t+1}(z)$ is the learned RPM component constructed from the RWGOP.

Figure 1 compares the Sharpe ratios of $\overline{R}_{t+1}^{\mathrm{RPM}}(z)$ and $\overline{R}_{t+1}^{\mathrm{RF}}(z)$ at each value of the final prediction regularization parameter z, separately for each target universe. The figure is read within each panel: at a given value of z, the relevant comparison is whether the RPM curve lies above the RF curve in the same universe.

Figure 1 shows that RPM improves out-of-sample Sharpe ratios relative to RF in the All universe over a broad range of regularization values. At very low z, the two strategies are nearly identical, with Sharpe ratios around 3.6 to 3.7. As z increases, however, the RPM curve remains above the RF curve and the gap widens materially. Around $z = 10^0$, the gain is roughly one-half Sharpe point, and the advantage remains large at $z = 10^1$ and $z = 10^2$. The learned RWGOP therefore improves performance relative to the baseline RF benchmark in the broad cross section.

This improvement is robust across size-sorted universes. In Mega and Large, RPM is consistently above RF with modest but persistent gains across the grid. In Small and Micro, the gains are larger, especially at moderate and high values of z, where RPM declines more slowly than RF. The same learned RWGOP is estimated once on the full "All" universe and then carried into each target universe, so these results show that the learned characteristic geometry transfers successfully across market-capitalization groups.

Performance with real-time selected z. We next evaluate whether the RPM gains survive an implementable real-time choice of the final prediction regularization parameter. For each model $m \in \{\mathrm{RPM}, \mathrm{RF}, \mathrm{LR}\}$, let $\overline{R}_{t+1}^{m}(z)$ denote the fixed-z return series defined above. At each month t, model m selects its regularization parameter using only realized out-of-sample returns available prior to month t. Specifically, among the candidate values from the benchmark z-grid in (24) that have accumulated at least 12 historical out-of-sample observations, we choose the value of z with the highest realized Sharpe ratio over the expanding history up to month $t-1$:

$$\hat{z}_t^{m} = \underset{z \in \mathcal{Z}_t}{\operatorname{argmax}}\, \widehat{\mathrm{SR}}_{t-1}^{m}(z),$$

where $\mathcal{Z}_t$ denotes the subset of the benchmark z-grid in (24) that is eligible at month t. The selected-z return is then

$$\overline{R}_{t+1}^{m,\mathrm{sel}} = \overline{R}_{t+1}^{m}(\hat{z}_t^{m}), \tag{37}$$

Because each candidate z must first accumulate 12 out-of-sample observations, the selected-z sample begins in January 1994.

We then test whether the selected-z RPM strategy delivers returns not spanned by the

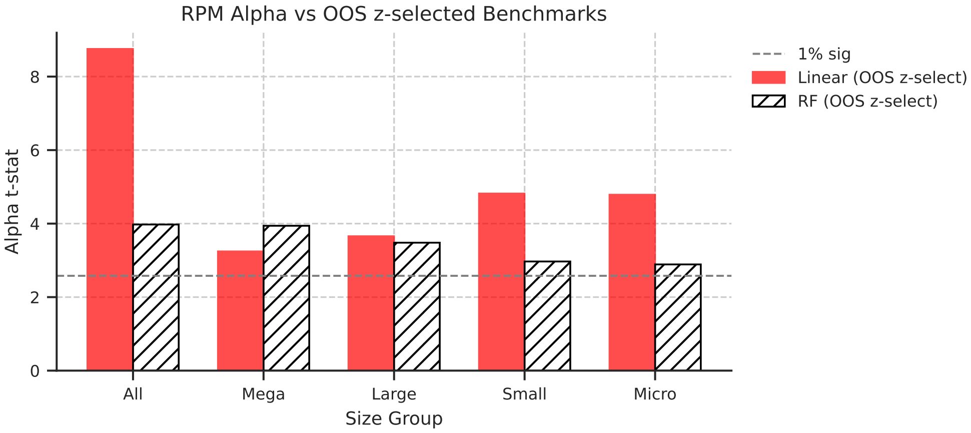

Figure 2: Alpha t-stat of the Real-Time Selected-z RPM Strategy Relative to Real-Time Selected-z Benchmark Ensembles

Note: The figure reports alpha t-statistics from time-series regressions of the real-time selected-z RPM strategy $\overline{R}_{t+1}^{\text{RPM,sel}}$ defined in (37) on the corresponding real-time selected-z benchmark strategies $\overline{R}_{t+1}^{\text{LR,sel}}$ and $\overline{R}_{t+1}^{\text{RF,sel}}$, estimated separately for each evaluation universe. At each month, each model selects its prediction regularization parameter from the benchmark z-grid in (24) using only realized out-of-sample returns observed up to the previous month, as described in (37). Alpha t-statistics are computed using heteroskedasticity- and autocorrelation-consistent standard errors with proper lags. The dashed horizontal line marks the 1% significance threshold. The out-of-sample period is from 1994 to 2024.

corresponding selected-z benchmark strategies. For $b \in \{\text{RF}, \text{LR}\}$, we estimate

$$\overline{R}_{t+1}^{\text{RPM,sel}} = \alpha^{\text{RPM}|b} + \beta^{\text{RPM}|b}\,\overline{R}_{t+1}^{b,\text{sel}} + \varepsilon_{t+1}, \tag{38}$$

and report the associated alpha t-statistics by target universe.

Figure 2 shows that the selected-z RPM strategy delivers economically and statistically meaningful alpha in the All universe. Relative to the real-time selected-z LR benchmark, the alpha t-statistic is close to 9, and relative to the real-time selected-z RF benchmark, it is about 4. These magnitudes are well above the 1% significance threshold and show that the RPM gains are not an artifact of fixing z ex ante.

These gains are robust across the size-sorted universes. In Mega and Large, the alpha t-statistics are around 3 to 4 relative to both selected-z benchmarks. In Small and Micro, the alpha t-statistics are larger: relative to LR they are close to 5, and relative to RF they are around 3. Thus, the selected-z RPM strategy continues to outperform both benchmarks across all market-capitalization groups. These alpha results show that the learned RWGOP captures pricing structure not spanned by either benchmark model.

5.5 Principal Pricing Directions

This subsection studies whether the learned RWGOP is effectively low-dimensional and whether a small number of principal pricing directions retains most of RPM's predictive content. As in the previous subsection, we evaluate the portfolio implications of these principal pricing directions in two ways: first, by studying performance as a function of the prediction regularization parameter z, and second, by studying performance under real-time selected z. We begin by characterizing the dimensionality of the learned RWGOP itself.

For each out-of-sample month t and Stage 1 specification (z_M, T_u, T_M), let

$$M_{1,t}(z_M, T_u, T_M) = V_t(z_M, T_u, T_M)\, \Lambda_t(z_M, T_u, T_M)\, V_t(z_M, T_u, T_M)' \tag{39}$$

denote the eigenvalue decomposition of the learned RWGOP, where

$$\Lambda_t(z_M, T_u, T_M) = \operatorname{diag}(\lambda_{t,1}, \ldots, \lambda_{t,d})$$

contains the eigenvalues ordered from largest to smallest.

To summarize the dimensionality of the learned RWGOP over time, we compute its entropy-based effective rank. For a positive semidefinite matrix with eigenvalues $\lambda_{t,1}, \ldots, \lambda_{t,d}$, define

$$p_{t,j} = \frac{\lambda_{t,j}}{\sum_{\ell=1}^{d} \lambda_{t,\ell}}, \qquad \operatorname{erank}(M_{1,t}) = \exp\left(-\sum_{j=1}^{d} p_{t,j} \log p_{t,j} \right).$$

This statistic equals one when the spectrum is concentrated in a single direction and increases as the eigenvalue mass becomes more diffuse. In the empirical implementation, we compute effective rank separately for each learned RWGOP and then average across retained specifications date by date.

Figure 3 shows that the learned RWGOP is economically low-dimensional even though the original characteristic space is much larger. The figure also shows that the effective rank depends on the metric-construction window T_M, so the breadth of the learned characteristic geometry varies systematically with the horizon over which the monthly RWGOP objects are aggregated.

We next ask whether this low-dimensional structure is economically relevant for portfolio performance. For each date t and Stage 1 specification (z_M, T_u, T_M), define the rank-κ

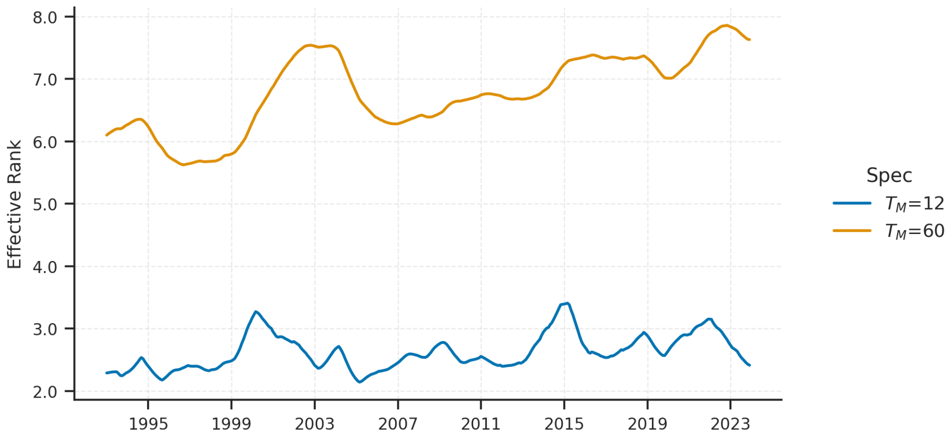

Figure 3: Effective Rank of the Learned RWGOP

Note: The figure reports the 12-month rolling average of the entropy-based effective rank of the learned RWGOP $M_{1,t}(z_M, T_u, T_M)$ defined in (39), estimated on the All universe. The plotted lines correspond to the two values of the Stage 1 metric-construction window T_M. Higher values indicate a more diffuse eigenvalue spectrum, while lower values indicate that the learned characteristic geometry is concentrated in fewer directions. The out-of-sample period is from 1993 to 2024.

approximation

$$M_{1,t}^{(\kappa)}(z_M, T_u, T_M) = V_{t,\kappa}(z_M, T_u, T_M)\, \Lambda_{t,\kappa}(z_M, T_u, T_M)\, V_{t,\kappa}(z_M, T_u, T_M)', \tag{40}$$

where $V_{t,\kappa}$ contains the top κ eigenvectors and $\Lambda_{t,\kappa}$ the corresponding eigenvalues. Replacing $M_{1,t}$ with $M_{1,t}^{(\kappa)}$ in the Stage 2 forecasting step defined in Section 5 yields rank-restricted RPM returns

$$R_{t+1}^{\text{RPM},1}(z_M, T_u, T_M, T, z, \kappa).$$

As in (35), the reported rank-restricted RPM strategy combines the RF benchmark with the corresponding learned RPM component:

$$\overline{R}_{t+1}^{\text{RPM}}(z, \kappa) = \omega_t^{\text{RPM},0}(z)\, \overline{R}_{t+1}^{\text{RF}}(z) + \omega_t^{\text{RPM},1}(z, \kappa)\, \overline{R}_{t+1}^{\text{RPM},1}(z, \kappa), \tag{41}$$

where $\kappa \in \{1, 2, 5\}$. We refer to the unreduced model as the full-rank specification.

Low-rank versions of RPM retain a substantial share of the performance of the full-rank model as a function of z. Figure 4 reports the Sharpe ratio of $\overline{R}_{t+1}^{\text{RPM}}(z, \kappa)$ for $\kappa \in \{1, 2, 5\}$, together with the full-rank specification. The rank-5 specification is often very close to the full-rank model, while the rank-1 and rank-2 specifications remain economically competitive

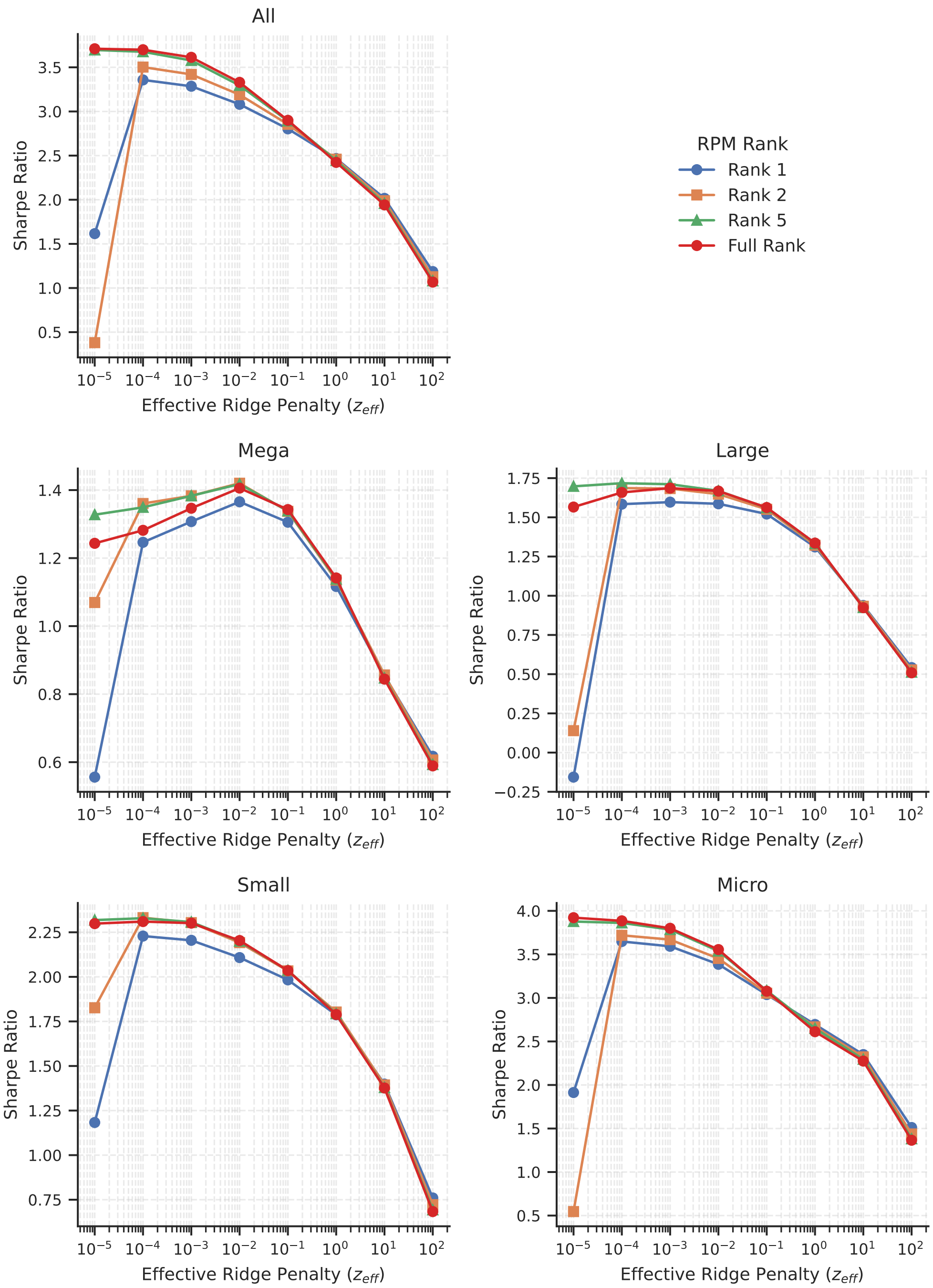

Figure 4: Sharpe Ratio of Rank-Restricted RPM as a Function of z

Note: The figure reports the out-of-sample Sharpe ratio of the rank-restricted RPM strategy $\overline{R}^{\text{RPM}}_{t+1}(z, \kappa)$ defined in (41), plotted as a function of the prediction regularization parameter z in (24) and separately by evaluation universe. The plotted lines correspond to rank 1, rank 2, rank 5, and the full-rank specification. The out-of-sample period is from 1993 to 2024.

in many universes. This pattern is consistent with the low-dimensional structure suggested by Figure 3.

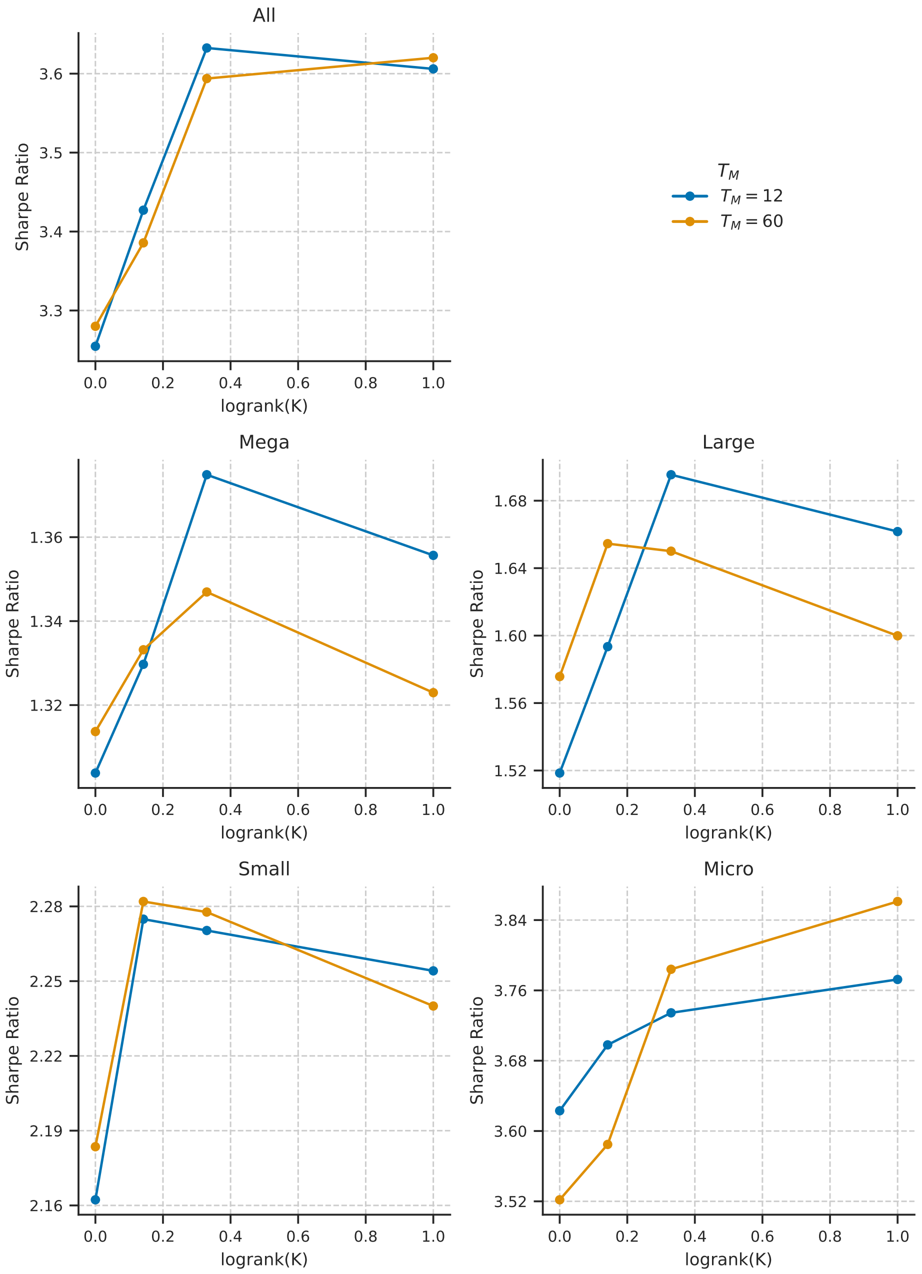

Figure 5: Sharpe Ratio of Rank-Restricted RPM with Real-Time Selected z

Note: The figure reports the out-of-sample Sharpe ratio of the real-time selected-z rank-restricted RPM strategy $\overline{R}_{t+1}^{\text{RPM,sel}}(\kappa, T_M)$ defined in (42). At each month, each rank restriction and metric-construction window selects its prediction regularization parameter using the real-time selected-z procedure in (37). The plotted lines correspond to $T_M \in \{12, 60\}$, and the interior points correspond to $\kappa \in \{1, 2, 5\}$, while the right-most point reports the corresponding full-rank specification. The out-of-sample period is from 1994 to 2024.

This low-rank performance survives real-time implementation. Using the same real-time selected-z procedure defined in (37), we construct rank-restricted selected-z RPM strategies for each κ and T_M:

$$\overline{R}^{\mathrm{RPM,sel}}_{t+1}(\kappa, T_M) = \overline{R}^{\mathrm{RPM}}_{t+1}(\hat{z}_t^{\kappa,T_M}, \kappa, T_M), \tag{42}$$

where $\hat{z}_t^{\kappa,T_M}$ is chosen month by month using only prior out-of-sample returns. Figure 5 reports the Sharpe ratios of these selected-z rank-restricted RPM strategies across rank restrictions $\kappa \in \{1, 2, 5\}$ and metric windows $T_M \in \{12, 60\}$. The figure shows that low-rank restrictions continue to preserve a substantial share of RPM's predictive content even when each rank-restricted strategy selects its own prediction regularization parameter using only past out-of-sample performance.

5.6 RPM as a Pseudo-Linear Factor Model

This subsection studies whether the leading principal pricing direction of the learned RWGOP admits a pseudo-linear interpretation and whether it aligns closely with the loading vector of the LR benchmark. Building on the principal pricing directions studied in the previous subsection, the key idea is that, when the learned RWGOP is restricted to rank one, the learned RPM component depends on characteristics only through a single linear index passed through a nonlinear transformation. We then test this interpretation empirically by comparing the leading principal pricing directions of the learned RWGOP to the loading vector of the LR benchmark.

Recall that the LR benchmark takes the form

$$\pi_t^{\mathrm{LR}}(x; T, z) = x'\theta_t(T, z), \tag{43}$$

where $\theta_t(T, z)$ is the rolling-window linear loading vector. We say that a portfolio rule is *pseudo-linear* if it can be written as

$$\pi_t(x) = h_t(x'\vartheta_t), \tag{44}$$

for some loading vector $\vartheta_t \in \mathbb{R}^d$ and some scalar nonlinear function $h_t : \mathbb{R} \to \mathbb{R}$.

Now consider the rank-one restriction of the learned RWGOP:

$$M^{(1)}_{1,t}(z_M, T_u, T_M) = \lambda_{t,1}(z_M, T_u, T_M)\, v_{t,1}(z_M, T_u, T_M)\, v_{t,1}(z_M, T_u, T_M)'. \tag{45}$$

Its square-root transform satisfies

$$\left(M^{(1)}_{1,t}(z_M, T_u, T_M)\right)^{1/2} x = \lambda_{t,1}(z_M, T_u, T_M)^{1/2}\, v_{t,1}(z_M, T_u, T_M)\, v_{t,1}(z_M, T_u, T_M)' x. \tag{46}$$

Thus, the transformed characteristic vector depends on x only through the scalar projection

$$v_{t,1}(z_M, T_u, T_M)'x.$$

It follows that the corresponding rank-one learned RPM component can be written in pseudo-linear form,

$$\pi_t^{\mathrm{RPM},1}(x; z_M, T_u, T_M, T, z, \kappa = 1) = h_t(v_{t,1}(z_M, T_u, T_M)'x)\,, \tag{47}$$

for some time-varying nonlinear function h_t. Appendix D.1 provides the formal derivation. In this sense, the leading principal pricing direction plays the same role as the loading vector in a linear factor model, except that the final pricing rule remains nonlinear in that one-dimensional index.

This pseudo-linear interpretation implies that the first principal pricing direction should be more closely aligned with the LR loading vector than the higher-order directions. We study this comparison on the All universe, which is the universe used to estimate the learned RWGOP in Section 5.

For the LR benchmark, define the window-averaged loading vector

$$\bar{\theta}_t(z) = \frac{1}{|W|} \sum_{T \in \{12,60,120,240,360\}} \theta_t(T, z). \tag{48}$$

We then evaluate this direction at the real-time selected LR regularization parameter,

$$\hat{z}_t^{\mathrm{LR}} \in \arg\max_z \widehat{\mathrm{SR}}_{t-1}^{\mathrm{LR}}(z), \tag{49}$$

using the same selected-z procedure as in (37). The LR direction entering the comparison is therefore

$$\bar{\theta}_t\big(\hat{z}_t^{\mathrm{LR}}\big).$$

For the learned RWGOP $M_{1,t}(z_M, T_u, T_M)$, let $v_{t,k}(z_M, T_u, T_M)$ denote its k-th eigenvector. Because eigenvectors are defined only up to sign, we first align their signs within each (t, k) cell and then average them across the retained Stage 1 specifications. Letting $\mathcal{U}$ denote the retained specification set, we define the sign-aligned average principal pricing direction

$$\bar{v}_{t,k} = \frac{1}{|\mathcal{U}|} \sum_{(z_M, T_u, T_M) \in \mathcal{U}} s_{t,k}(z_M, T_u, T_M)\, v_{t,k}(z_M, T_u, T_M), \tag{50}$$

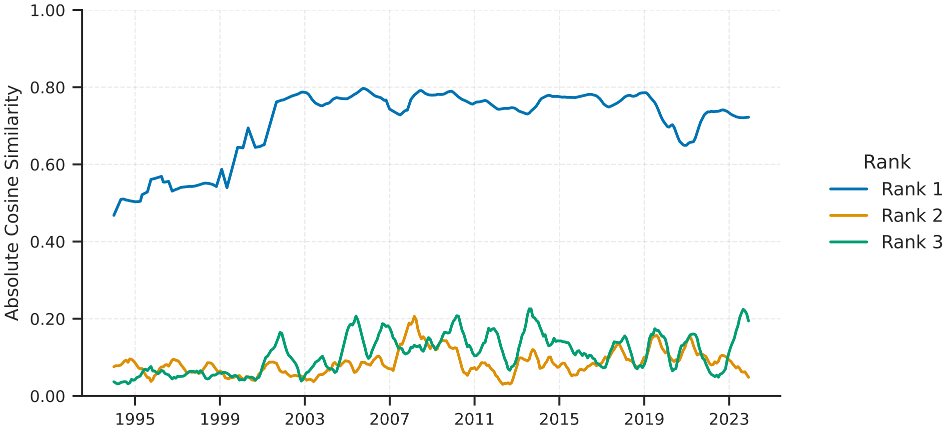

Figure 6: Alignment Between the Averaged LR Loading and Principal Pricing Directions

Note: The figure reports the 12-month rolling average of the absolute cosine similarity $\mathrm{CS}_{t,k}$ defined in (51) between the averaged All-universe LR loading $\bar{\theta}_t(\hat{z}_t^{\mathrm{LR}})$ and the sign-aligned average principal pricing directions $\bar{v}_{t,k}$ defined in (48), (49), and (50). Higher values indicate greater alignment between the corresponding linear benchmark direction and principal pricing direction. The out-of-sample period is from 1994 to 2024.

where $s_{t,k}(z_M, T_u, T_M) \in \{-1, 1\}$ aligns each eigenvector with a common reference direction before averaging.

We measure alignment by the absolute cosine similarity

$$\mathrm{CS}_{t,k} = \frac{\left|\bar{\theta}_t\left(\hat{z}_t^{\mathrm{LR}}\right)' \bar{v}_{t,k}\right|}{\left\|\bar{\theta}_t\left(\hat{z}_t^{\mathrm{LR}}\right)\right\| \, \left\|\bar{v}_{t,k}\right\|}. \tag{51}$$

To reduce short-run noise, we report a 12-month rolling average of $\mathrm{CS}_{t,k}$ over the selected-z sample.

The first principal pricing direction is much more closely aligned with the LR benchmark than the higher-order directions. Figure 6 reports the rolling average of the absolute cosine similarity between the real-time selected, window-averaged LR loading and the sign-aligned average principal pricing directions. The first principal pricing direction exhibits the strongest alignment throughout the sample, whereas the second and third directions are substantially less aligned. This pattern is consistent with the pseudo-linear interpretation of rank-one RPM and suggests that higher-order principal pricing directions capture pricing structure that is not well described by the linear benchmark.

6 Conclusion

This paper studies how to learn the characteristic directions that matter for pricing. We show that this problem can be framed as learning both a portfolio rule and the low-dimensional subspace of characteristic space on which that rule depends. Our theory identifies the RWGOP as the central object for this task, and that result motivates the RPM. The RPM uses the sample RWGOP to learn the pricing-relevant geometry of characteristic space and then uses that geometry to construct the SDF within a closed-form large factor model. Empirically, the RPM improves out-of-sample performance relative to large-factor benchmarks based on fixed random features, while low-rank versions of the model retain much of the performance of the full-rank specification.

A natural next step is to better understand the economic content of the principal pricing directions learned by the model. It would also be valuable to extend the framework to other asset classes and to richer recursive learning schemes. More broadly, the paper suggests that recursive feature learning offers a promising way to combine flexibility, tractability, and interpretability in empirical asset pricing.

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A Appendix

This appendix follows the structure of the paper. Section B collects technical material related to the framework and kernel representation in Section 3. Section C contains the proofs and supporting results for the RWGOP and principal pricing directions in Section 4. Section D provides the appendix material referenced in the empirical section, including the pseudo-linear interpretation of the rank-one RPM. Section E describes the numerical implementation.

B Proofs for Section 3: Framework and Kernel Representation

B.1 Proof of Theorem 2: Arc-Cosine Kernel

Proof of Theorem 2. Let

$$u = w'x, \qquad v = w'y.$$

Since $w \sim \mathcal{N}(0, I_d)$, the vector (u, v) is jointly Gaussian with

$$\mathrm{Var}(u) = \|x\|^2, \quad \mathrm{Var}(v) = \|y\|^2, \quad \mathrm{Cov}(u, v) = x'y.$$

Write $\|x\| = a$, $\|y\| = b$, and define

$$\rho = \frac{x'y}{\|x\|\|y\|} = \cos\theta, \qquad \theta \in [0, \pi].$$

Then we may represent

$$u = aZ_1, \qquad v = b\big(\rho Z_1 + \sqrt{1-\rho^2}\, Z_2\big),$$

where $Z_1, Z_2 \overset{iid}{\sim} \mathcal{N}(0, 1)$. Hence

$$K(x, y) = ab\,\mathbb{E}\Big[(Z_1)_+\big(\rho Z_1 + \sqrt{1-\rho^2}\, Z_2\big)_+\Big],$$

where $(t)_+ = \max(0, t)$.

Introduce polar coordinates $Z_1 = r\cos\varphi$, $Z_2 = r\sin\varphi$. Using $\rho = \cos\theta$ and $\sqrt{1-\rho^2} = \sin\theta$, we have

$$\rho Z_1 + \sqrt{1-\rho^2} Z_2 = r\cos(\varphi - \theta).$$

With the standard Gaussian density $(2\pi)^{-1}e^{-r^2/2}$, this yields

$$K(x,y) = ab \cdot \frac{1}{2\pi} \int_0^\infty \int_0^{2\pi} r^3 (\cos\varphi)_+ (\cos(\varphi-\theta))_+ e^{-r^2/2}\, d\varphi\, dr.$$

The radial integral evaluates to

$$\int_0^\infty r^3 e^{-r^2/2}\, dr = 2,$$

so

$$K(x,y) = ab \cdot \frac{1}{\pi} \int_0^{2\pi} (\cos\varphi)_+ (\cos(\varphi-\theta))_+\, d\varphi.$$

The integrand is nonzero only when both cosines are positive, which occurs on an interval of length $\pi - \theta$. A direct trigonometric calculation gives

$$\int_0^{2\pi} (\cos\varphi)_+ (\cos(\varphi-\theta))_+\, d\varphi = \frac{1}{2}\Big(\sin\theta + (\pi-\theta)\cos\theta\Big).$$

Substituting back completes the proof. □

B.2 Gradient of the Arc-Cosine Portfolio Kernel

The recursive update of the characteristic metric uses the gradient of the portfolio rule with respect to characteristics. For the arc-cosine kernel, this gradient admits a closed-form expression that makes transparent how the learned metric responds to local pricing sensitivities. The formula below separates a self-exposure term from a cross-stock similarity term, which is useful both computationally and economically.

Lemma 2 (Gradient of K_M) *Let M be symmetric positive semidefinite. The gradient of $K_M(x,y)$ with respect to x is*

$$\nabla_x K_M(x,y) = \frac{\sqrt{b}}{2\pi\sqrt{a}} \sin\theta\, Mx + \frac{\pi-\theta}{2\pi}\, My, \tag{52}$$

where $a = x'Mx$, $b = y'My$, and

$$\theta = \arccos\left(\frac{x'My}{\sqrt{ab}}\right).$$

Proof of Lemma 2. Write the kernel in modular form. Define

$$a(x) := x'Mx, \qquad b := y'My, \qquad c(x) := x'My,$$

and

$$s(x) := \sqrt{a(x)}, \qquad t := \sqrt{b}, \qquad \rho(x) := \frac{c(x)}{s(x)t}, \qquad \theta(x) := \arccos(\rho(x)).$$

Let

$$\Phi(\theta) := \sin\theta + (\pi - \theta)\cos\theta.$$

Then

$$K_M(x, y) = \frac{s(x)t}{2\pi}\,\Phi(\theta(x)). \tag{53}$$

Step 1: Basic derivatives. Since M is symmetric,

$$\nabla_x a(x) = 2Mx.$$

Hence

$$\nabla_x s(x) = \frac{Mx}{s(x)}. \tag{54}$$

Also,

$$\nabla_x c(x) = My, \tag{55}$$

and $t = \sqrt{b}$ is constant with respect to x.

Step 2: Derivative of $\Phi(\theta)$. Differentiate

$$\Phi'(\theta) = -(\pi - \theta)\sin\theta.$$

Therefore

$$\nabla_x \Phi(\theta(x)) = \Phi'(\theta(x))\,\nabla_x\theta(x) = -(\pi - \theta)\sin\theta\left(-\frac{1}{\sqrt{1-\rho^2}}\nabla_x\rho\right). \tag{56}$$

Since $\sin\theta = \sqrt{1-\rho^2}$,

$$\nabla_x \Phi(\theta(x)) = (\pi - \theta(x))\, \nabla_x \rho(x). \tag{57}$$

Step 3: Gradient of $\rho(x)$. Recall $\rho(x) = \dfrac{c(x)}{s(x)t}$. Since t is constant in x,

$$\nabla_x \rho(x) = \frac{1}{t}\nabla_x \left(\frac{c(x)}{s(x)}\right) = \frac{1}{t}\left(\frac{\nabla_x c(x)}{s(x)} - \frac{c(x)}{s(x)^2}\nabla_x s(x)\right).$$

Using (54) and (55),

$$\nabla_x \rho(x) = \frac{1}{st}My - \frac{\rho}{s^2}Mx. \tag{58}$$

Step 4: Assemble $\nabla_x K_M(x,y)$. Differentiate (53):

$$\nabla_x K_M(x,y) = \frac{t}{2\pi}\Big(\nabla_x s(x)\, \Phi(\theta(x)) + s(x)\, \nabla_x \Phi(\theta(x))\Big).$$

Substitute (54), (57), and (58):

$$\nabla_x K_M(x,y) = \frac{t}{2\pi}\left(\frac{Mx}{s}\Phi(\theta) + s(\pi-\theta)\left(\frac{1}{st}My - \frac{\rho}{s^2}Mx\right)\right).$$

This becomes

$$\nabla_x K_M(x,y) = \frac{t}{2\pi}\cdot\frac{\Phi(\theta) - (\pi-\theta)\rho}{s}Mx + \frac{\pi-\theta}{2\pi}My.$$

Since $\rho = \cos\theta$ and

$$\Phi(\theta) - (\pi-\theta)\rho = \sin\theta,$$

we obtain

$$\nabla_x K_M(x,y) = \frac{t}{2\pi}\cdot\frac{\sin\theta}{s}Mx + \frac{\pi-\theta}{2\pi}My.$$

Recalling $s = \sqrt{a}$ and $t = \sqrt{b}$ yields the claim. □

For the fitted portfolio rule

$$f_T(X_{i,\tau}) = \sum_{t=1}^{T}\sum_{j=1}^{N_t} K_M(X_{i,\tau}, X_{j,t-1})R_{j,t}\xi_t,$$

Lemma 2 implies

$$\begin{aligned}\nabla_{X_{i,\tau}} f_T(X_{i,\tau}) &= MX_{i,\tau}\sum_{t=1}^{T}\sum_{j=1}^{N_t}\frac{\sqrt{X'_{j,t-1}MX_{j,t-1}}}{2\pi\sqrt{X'_{i,\tau}MX_{i,\tau}}}\sin\theta_{i,\tau;j,t}\,R_{j,t}\xi_t \\ &\quad + M\sum_{t=1}^{T}\sum_{j=1}^{N_t}\frac{\pi-\theta_{i,\tau;j,t}}{2\pi}X_{j,t-1}R_{j,t}\xi_t,\end{aligned} \tag{59}$$

where

$$\theta_{i,\tau;j,t} = \arccos\left(\frac{X'_{i,\tau}MX_{j,t-1}}{\sqrt{X'_{i,\tau}MX_{i,\tau}}\sqrt{X'_{j,t-1}MX_{j,t-1}}}\right).$$

This decomposition clarifies that the gradient contains both a self-exposure component and a similarity-weighted cross-stock component.

C Proofs for Section 4: RWGOP and Principal Pricing Directions

C.1 Proof of Lemma 1

Lemma 1 shows that the RWGOP identifies the linear subspace through which the portfolio rule depends on characteristics. The proof proceeds by showing that directions in the null space of M_* must have zero directional derivative almost surely, and then using analyticity to extend that local statement globally. The converse rank bound follows from the chain rule.

Proof of Lemma 1. Step 1: Nullspace implies vanishing directional derivatives. Let $P_\perp := I_d - P$ be the orthogonal projector onto $\ker(M_*)$. For any $v \in \mathbb{R}^d$,

$$v'M_*v = E[(Jf(X)v)'\Sigma(X)(Jf(X)v)] \geq 0.$$

If $v \in \ker(M_*)$, then $v'M_*v = 0$ and hence

$$(Jf(X)v)'\Sigma(X)(Jf(X)v) = 0 \quad \text{a.s.}$$

By (18), this implies $\|Jf(X)v\|^2 = 0$ a.s., i.e.

$$Jf(X)v = 0 \qquad \text{a.s.} \tag{60}$$

In particular, for each fixed $v \in \ker(M_*)$, the real analytic function

$$h_v(x) \;:=\; Jf(x)v$$

vanishes for Lebesgue-a.e. $x \in \mathcal{S}$ because X has a density positive a.e. Since a nonzero real analytic function cannot vanish on a set of positive Lebesgue measure, we conclude that

$$Jf(x)v = 0 \qquad \text{for all } x \in \mathcal{S}, \text{ for all } v \in \ker(M_*). \tag{61}$$

Equivalently,

$$Jf(x)P_\perp = 0 \qquad \text{for all } x \in \mathcal{S}. \tag{62}$$

Step 2: f is constant along $\ker(M_*)$ directions. Fix $x \in \mathcal{S}$ and $w \in \ker(M_*)$ such that the segment $\{x + tw : t \in [0,1]\} \subset \mathcal{S}$. Define $\phi(t) := f(x + tw)$. Then ϕ is differentiable and by (61),

$$\phi'(t) = Jf(x + tw)w = 0 \qquad \forall t \in [0,1].$$

Hence $\phi(1) = \phi(0)$, i.e.

$$f(x + w) = f(x) \qquad \text{whenever } w \in \ker(M_*),\ x, x + w \in \mathcal{S}. \tag{63}$$

Step 3: Define g and show $f(x) = g(U'x)$. For $y \in U'\mathcal{S}$, choose any $x \in \mathcal{S}$ such that $U'x = y$ and define

$$g(y) \;:=\; f(x).$$

This is well-defined: if $x_1, x_2 \in \mathcal{S}$ satisfy $U'x_1 = U'x_2$, then $P(x_1 - x_2) = 0$, so $x_1 - x_2 \in \ker(M_*)$, and (63) implies $f(x_1) = f(x_2)$. Thus g is well-defined and satisfies $f(x) = g(U'x)$.

To see that g is real analytic, note that after an orthogonal change of coordinates one may write $x = (U\ U_\perp)\binom{y}{z}$ with $y = U'x$ and $z = U_\perp' x$. Then (62) implies $\partial f/\partial z \equiv 0$, hence in these coordinates f depends only on y; analyticity of g follows from analyticity of f.

This proves part (1).

Step 4: Converse rank bound. Suppose $f(x) = g(B'x)$ with $B \in \mathbb{R}^{d\times r}$ having orthonormal columns. Then by the chain rule,

$$Jf(x) = Jg(B'x)\, B'.$$

Therefore

$$Jf(x)'\Sigma(X)Jf(x) = B\left(Jg(B'x)'\Sigma(X)Jg(B'x)\right)B',$$

whose rank is at most r for each x. Taking expectation yields $\text{rank}(M_*) \leq r$. This proves part (2).

Step 5: Full rank under non-degeneracy. Under the assumptions of part (3), for any $a \neq 0$,

$$a'\,E\big[Jg(B'X)'\Sigma(X)Jg(B'X)\big]\,a = E\big[(Jg(B'X)a)'\Sigma(X)(Jg(B'X)a)\big].$$

By the stated non-degeneracy, the integrand is > 0 with positive probability, hence the expectation is strictly positive. Therefore

$$E\big[Jg(B'X)'\Sigma(X)Jg(B'X)\big] \succ 0,$$

and so

$$M_* = B\,E\big[Jg(B'X)'\Sigma(X)Jg(B'X)\big]\,B'$$

has rank r. This proves part (3). □

C.2 Approximation Error from Principal Pricing Directions

Corollary 3 gives the eigenvalue spectrum of M_* a direct approximation interpretation. The relevant error is the expected conditional second moment of the payoff loss from replacing the full portfolio rule by its projection onto the span of the leading principal pricing directions. The proof uses a matrix-normal projection argument. We first control the projection error under a Gaussian design, then specialize that bound to the row-wise portfolio-rule structure used in the paper, and finally identify the resulting projected gradient energy with the trailing eigenvalue mass of the RWGOP.

Throughout this subsection, we distinguish between the row-wise Jacobian

$$Jf(X) \in \mathbb{R}^{N\times d}$$

and the full Jacobian with respect to the vectorized matrix,

$$J_{\text{vec}(X)}f(X) \in \mathbb{R}^{N\times(Nd)}.$$

Lemma 3 (Gaussian Poincaré / Brascamp and Lieb 1976 inequality) *Let $Z \sim N(0, C)$ be a Gaussian vector in $\mathbb{R}^m$ with covariance matrix $C \in \mathbb{R}^{m\times m}$ symmetric positive definite. Let $h : \mathbb{R}^m \to \mathbb{R}^k$ be weakly differentiable with $E\|Jh(Z)\|_F^2 < \infty$, where $Jh(z) \in \mathbb{R}^{k\times m}$ denotes the Jacobian matrix.*

Then

$$E\|h(Z) - E[h(Z)]\|^2 \;\le\; E\big[\mathrm{tr}\big(Jh(Z)\, C\, Jh(Z)'\big)\big]\,. \tag{64}$$

Proof. Write $h = (h_1, \ldots, h_k)'$. Then

$$E\|h(Z) - E[h(Z)]\|^2 = \sum_{i=1}^{k} \mathrm{Var}(h_i(Z)).$$

For each component h_i, the scalar Gaussian Poincaré inequality of Brascamp and Lieb (1976) gives

$$\mathrm{Var}(h_i(Z)) \le E[\nabla h_i(Z)' C\, \nabla h_i(Z)]\,.$$

Summing over $i = 1, \ldots, k$ yields

$$E\|h(Z) - E[h(Z)]\|^2 \le \sum_{i=1}^{k} E[\nabla h_i(Z)' C\, \nabla h_i(Z)]\,.$$

Since the ith row of $Jh(Z)$ equals $\nabla h_i(Z)'$, we obtain

$$\sum_{i=1}^{k} \nabla h_i(Z)' C\, \nabla h_i(Z) = \mathrm{tr}\big(Jh(Z)\, C\, Jh(Z)'\big),$$

which proves the result. □

Lemma 4 (Σ-weighted conditional Gaussian Poincaré bound) *Let $X \in \mathbb{R}^{N\times d}$ satisfy*

$$X \;=\; A^{1/2} Y B^{1/2}, \qquad \mathrm{vec}(Y) \sim N(0, I_{Nd}),$$

where $A \in \mathbb{R}^{N\times N}$ and $B \in \mathbb{R}^{d\times d}$ are symmetric positive definite. Equivalently,

$$\mathrm{vec}(X) \;\sim\; N\big(0,\; B \otimes A\big).$$

Let $f : \mathbb{R}^{N\times d} \to \mathbb{R}^N$ *be weakly differentiable and assume*

$$E\|f(X)\|^2 < \infty, \qquad E\|J_{\mathrm{vec}(X)} f(X)\|_F^2 < \infty,$$

where $J_{\mathrm{vec}(X)} f(X) \in \mathbb{R}^{N\times(Nd)}$ *denotes the Jacobian with respect to* $\mathrm{vec}(X)$*.*

Fix an orthogonal decomposition $[U_p\ U_{-p}] \in \mathbb{R}^{d\times d}$ *and define the column projector*

$$P_{-p} \ := \ U_{-p}U_{-p}' \in \mathbb{R}^{d\times d}.$$

Let $\Sigma \in \mathbb{R}^{N\times N}$ *be symmetric positive definite and define*

$$f_p(XU_p) \ := \ E\big[f(X) \,\big|\, XU_p\big]\,.$$

Let $U'BU$ *be block-partitioned as*

$$U'BU \ = \ \begin{pmatrix} B_{yy} & B_{yz} \\ B_{zy} & B_{zz} \end{pmatrix}, \qquad B_{yy} \in \mathbb{R}^{p\times p},\ B_{zz} \in \mathbb{R}^{(d-p)\times(d-p)},$$

and define the Schur complement

$$\hat{B}_z \ := \ B_{zz} - B_{zy}B_{yy}^{-1}B_{yz} \ \in \ \mathbb{R}^{(d-p)\times(d-p)}.$$

Then

$$E[(f(X) - f_p(XU_p))'\Sigma(f(X) - f_p(XU_p))] \le E\Big[\mathrm{tr}\Big(\Sigma\, J_{\mathrm{vec}(X)} f(X) \\ \Big((U_{-p}\hat{B}_z U_{-p}') \otimes A\Big)\, J_{\mathrm{vec}(X)} f(X)'\Big)\Big]\,. \tag{65}$$

Proof. **Step 1: Reduce the Σ-weighted loss to an L^2 loss.** Let $g(X) := \Sigma^{1/2} f(X)$. By linearity of conditional expectation,

$$E[g(X) \mid XU_p] = \Sigma^{1/2} E[f(X) \mid XU_p] = \Sigma^{1/2} f_p(XU_p),$$

hence

$$E[(f - f_p)'\Sigma(f - f_p)] = E\|g(X) - E[g(X) \mid XU_p]\|^2. \tag{66}$$

Step 2: Rotate columns and compute the conditional covariance. Define

$$y := XU_p \in \mathbb{R}^{N\times p}, \qquad z := XU_{-p} \in \mathbb{R}^{N\times(d-p)}.$$

Then $X = yU_p' + zU_{-p}'$ and, vectorizing,

$$\begin{pmatrix} \mathrm{vec}(y) \\ \mathrm{vec}(z) \end{pmatrix} = \begin{pmatrix} U_p' \otimes I_N \\ U_{-p}' \otimes I_N \end{pmatrix} \mathrm{vec}(X).$$

Since $\mathrm{vec}(X) \sim N(0, B \otimes A)$ and U is orthogonal,

$$\begin{pmatrix} \mathrm{vec}(y) \\ \mathrm{vec}(z) \end{pmatrix} \sim N(0,\ (U'BU) \otimes A)\,.$$

Thus the conditional covariance of $\mathrm{vec}(z) \mid y$ is

$$\mathrm{Cov}(\mathrm{vec}(z) \mid y) = (B_{zz} \otimes A) - (B_{zy} \otimes A)(B_{yy} \otimes A)^{-1}(B_{yz} \otimes A).$$

Using $(B_{yy} \otimes A)^{-1} = B_{yy}^{-1} \otimes A^{-1}$ and the mixed-product rule gives

$$\mathrm{Cov}(\mathrm{vec}(z) \mid y) \;=\; \hat{B}_z \otimes A. \tag{67}$$

Step 3: Apply conditional vector-valued Gaussian Poincaré. Fix y and define $H_y(z) := g(yU_p' + zU_{-p}')$ as a function of z. Then $E[H_y(z) \mid y] = E[g(X) \mid y] = E[g(X) \mid XU_p]$ and, by Lemma 3 applied conditionally to $\mathrm{vec}(z) \mid y$ with covariance $\hat{B}_z \otimes A$,

$$E\big(\|H_y(z) - E[H_y(z) \mid y]\|^2 \,\big|\, y\big) \le E\Big(\mathrm{tr}\big(J_z H_y(z)\,(\hat{B}_z \otimes A)\,J_z H_y(z)'\big) \,\Big|\, y\Big)\,. \tag{68}$$

Taking expectations over y yields

$$E\|g(X) - E[g(X) \mid XU_p]\|^2 \le E\Big[\mathrm{tr}\big(J_z H_y(z)\,(\hat{B}_z \otimes A)\,J_z H_y(z)'\big)\Big]\,. \tag{69}$$

Step 4: Chain rule and appearance of $(U_{-p}\hat{B}_z U_{-p}') \otimes A$. From

$$\mathrm{vec}(X) = (U_p \otimes I_N)\,\mathrm{vec}(y) + (U_{-p} \otimes I_N)\,\mathrm{vec}(z),$$

we have

$$\frac{\partial\ \mathrm{vec}(X)}{\partial\ \mathrm{vec}(z)} \;=\; U_{-p} \otimes I_N.$$

Hence, by the chain rule,

$$J_z H_y(z) \;=\; J_{\mathrm{vec}(X)} g(X)\,(U_{-p} \otimes I_N).$$

Substituting into (69) gives

$$\operatorname{tr}\big(J_z H_y\,(\hat{B}_z \otimes A)\,J_z H_y'\big) = \operatorname{tr}\Big(J_{\mathrm{vec}(X)} g\,(U_{-p} \otimes I_N)\,(\hat{B}_z \otimes A)\,(U_{-p}' \otimes I_N)\,J_{\mathrm{vec}(X)} g'\Big).$$

Using

$$(U_{-p} \otimes I_N)(\hat{B}_z \otimes A)(U_{-p}' \otimes I_N) = (U_{-p}\hat{B}_z U_{-p}') \otimes A,$$

we obtain

$$\operatorname{tr}\big(J_z H_y\,(\hat{B}_z \otimes A)\,J_z H_y'\big) = \operatorname{tr}\Big(J_{\mathrm{vec}(X)} g\,\big((U_{-p}\hat{B}_z U_{-p}') \otimes A\big)\,J_{\mathrm{vec}(X)} g'\Big).$$

Step 5: Substitute back $g = \Sigma^{1/2} f$. Since $J_{\mathrm{vec}(X)} g = \Sigma^{1/2} J_{\mathrm{vec}(X)} f$,

$$\operatorname{tr}\Big(J_{\mathrm{vec}(X)} g\, M\, J_{\mathrm{vec}(X)} g'\Big) = \operatorname{tr}\Big(\Sigma\, J_{\mathrm{vec}(X)} f\, M\, J_{\mathrm{vec}(X)} f'\Big),$$

with $M = ((U_{-p}\hat{B}_z U_{-p}') \otimes A)$. Combine with (66) to obtain (65). □

Corollary 4 (Explicit bound using $\lambda_1(B)$) *Under the assumptions of Lemma 4,*

$$0 \preceq \hat{B}_z \preceq \lambda_1(B) I_{d-p} \quad \Rightarrow \quad U_{-p}\hat{B}_z U_{-p}' \preceq \lambda_1(B)\, P_{-p}.$$

Hence

$$E[(f - f_p)'\Sigma(f - f_p)] \ \leq\ \lambda_1(B)\, E\Big[\operatorname{tr}\Big(\Sigma\, J_{\mathrm{vec}(X)} f(X)\,\big(P_{-p} \otimes A\big)\,J_{\mathrm{vec}(X)} f(X)'\Big)\Big].$$

Proof. Since $B \succ 0$, every Schur complement of a principal block is positive semidefinite, so

$$0 \preceq \hat{B}_z.$$

Also, because $\hat{B}_z \preceq B_{zz}$ and $B_{zz} \preceq \lambda_1(B) I_{d-p}$,

$$\hat{B}_z \preceq \lambda_1(B) I_{d-p}.$$

Conjugating by U_{-p} yields

$$U_{-p}\hat{B}_z U_{-p}' \preceq \lambda_1(B)\, U_{-p} U_{-p}' = \lambda_1(B)\, P_{-p}.$$

Substituting this bound into (65) gives the result. □

Corollary 5 (Separable row-wise map, column projection) *Suppose*

$$f(X) = (f(x_i))_{i=1}^N, \qquad f : \mathbb{R}^d \to \mathbb{R}$$

is weakly differentiable with $E\|\nabla f(X_i)\|^2 < \infty$. *Define*

$$Jf(X) \in \mathbb{R}^{N \times d}, \qquad (Jf(X))_{i,:} \;:=\; \nabla f(X_i)'.$$

Let $S := U_{-p}\hat{B}_z U_{-p}' \in \mathbb{R}^{d \times d}$. *Then the right-hand side trace in* (65) *equals*

$$\operatorname{tr}\Big(\Sigma\, J_{\mathrm{vec}(X)} f(X)\, (S \otimes A)\, J_{\mathrm{vec}(X)} f(X)'\Big) \;=\; \operatorname{tr}\Big(\Sigma \left(A \odot (Jf(X)\, S\, Jf(X)')\right)\Big), \tag{70}$$

and therefore

$$E[(f(X) - f_p(XU_p))'\Sigma(f(X) - f_p(XU_p))] \;\le\; E\Big[\operatorname{tr}\Big(\Sigma \left(A \odot (Jf(X)\, S\, Jf(X)')\right)\Big)\Big].$$

In particular, using $S \preceq \lambda_1(B)P_{-p}$ *gives*

$$E[(f - f_p)'\Sigma(f - f_p)] \;\le\; \lambda_1(B)\, E\Big[\operatorname{tr}\Big(\Sigma \left(A \odot (Jf(X)\, P_{-p}\, Jf(X)')\right)\Big)\Big].$$

Similarly, since $A \preceq \lambda_1(A) I_N$, *we have*

$$A \odot C \preceq \lambda_1(A)\, C$$

for any symmetric positive semidefinite matrix C, *and hence*

$$E[(f(X) - f_p(XU_p))'\Sigma(f(X) - f_p(XU_p))] \;\le\; \lambda_1(A)\lambda_1(B)\, E\Big[\operatorname{tr}\Big(\Sigma\, Jf(X)\, P_{-p}\, Jf(X)'\Big)\Big].$$

Proof. For the ith component $f_i(X) = f(X_i)$, the gradient with respect to $\mathrm{vec}(X)$ is

$$\nabla_{\mathrm{vec}(X)} f_i(X) = e_i \otimes \nabla f(X_i),$$

hence $J_{\mathrm{vec}(X)} f(X)$ is block-diagonal with ith block row equal to $\nabla f(X_i)'$. A direct computation shows that

$$\left(J_{\mathrm{vec}(X)} f\, (S \otimes A)\, J_{\mathrm{vec}(X)} f'\right)_{ij} = A_{ij}\, \nabla f(X_i)' S \nabla f(X_j),$$

so

$$J_{\mathrm{vec}(X)}f\,(S\otimes A)\,J_{\mathrm{vec}(X)}f' = A\odot(Jf(X)SJf(X)').$$

Taking $\mathrm{tr}(\Sigma\,\cdot)$ yields (70). The two displayed bounds follow by substituting the operator inequalities for S and A. □

C.3 Proof of Corollary 3

Proof of Corollary 3. The result follows from the projection bounds proved above.

Step 1: Apply the row-wise column-projection bound. Under the matrix-normal design

$$X = A^{1/2}YB^{1/2}, \qquad \mathrm{vec}(Y)\sim N(0, I_{Nd}),$$

and the row-wise representation

$$f(X) = (f(x_i))_{i=1}^{N},$$

Corollary 5 gives

$$E[(f(X)-f_p(XU_p))'\Sigma(f(X)-f_p(XU_p))] \le \lambda_1(A)\lambda_1(B)\,E\Big[\mathrm{tr}\Big(\Sigma\,Jf(X)\,P_{-p}\,Jf(X)'\Big)\Big], \tag{71}$$

where

$$P_{-p} := U_{-p}U_{-p}'$$

is the orthogonal projector onto the span of the trailing $d-p$ eigenvectors of M_*.

Step 2: Identify the projected gradient energy with the spectral tail of M_*. Using cyclicity of the trace,

$$\mathrm{tr}\Big(\Sigma\,Jf(X)\,P_{-p}\,Jf(X)'\Big) = \mathrm{tr}\Big(P_{-p}\,Jf(X)'\Sigma\,Jf(X)\Big).$$

Taking expectations and using the definition of the RWGOP,

$$M_* \;=\; E[Jf(X)'(\Sigma + E[R|X]E[R|X]')\,Jf(X)] \;\succeq\; \tilde{M}_* = E[Jf(X)'\Sigma\,Jf(X)]\,,$$

we obtain

$$E\Big[\mathrm{tr}\Big(\Sigma\, Jf(X)\, P_{-p}\, Jf(X)'\Big)\Big] = \mathrm{tr}\Big(P_{-p}\, E[Jf(X)'\Sigma\, Jf(X)]\Big) = \mathrm{tr}(P_{-p}\tilde{M}_*) \;\leq\; \mathrm{tr}(P_{-p}M_*)$$

Now let

$$M_* = UDU', \qquad U = [U_p\; U_{-p}],$$

with eigenvalues ordered as

$$D = \mathrm{diag}(\lambda_1(M_*), \ldots, \lambda_d(M_*)).$$

Because $P_{-p} = U_{-p}U_{-p}'$ is the projector onto the trailing eigenspace,

$$\mathrm{tr}(P_{-p}M_*) = \sum_{k=p+1}^{d} \lambda_k(M_*) = \Lambda_{-p}(M_*).$$

Step 3: Conclude. Substituting this identity into (71) yields

$$E[(f(X) - f_p(XU_p))'\Sigma(f(X) - f_p(XU_p))] \leq \lambda_1(A)\lambda_1(B)\,\Lambda_{-p}(M_*),$$

which is exactly (20). □

D Proofs for Section 5: Empirical Interpretation

D.1 Pseudo-Linear Interpretation of the Rank-One RPM

This subsection formalizes the claim used in Section 5 that a rank-one learned RWGOP induces a pseudo-linear RPM component. The key observation is that once the metric has rank one, the kernel depends on characteristics only through a single scalar projection.

Lemma 5 *If $M = \lambda_1 v_1 v_1'$ has rank one, then*

$$K_M(x, y) = \lambda_1\, g(v_1'x, v_1'y), \tag{72}$$

where

$$g(a, b) = \frac{1}{2}\max(ab, 0) = \frac{1}{2}|b|\big(\max(a, 0)\mathbf{1}_{b>0} + \max(-a, 0)\mathbf{1}_{b<0}\big). \tag{73}$$

Proof of Lemma 5. For $a, b \in \mathbb{R}$ define

$$g(a,b) \;=\; \frac{|a|\,|b|}{2\pi}\Big(\sin\arccos(\operatorname{sign}(ab)) + (\pi - \arccos(\operatorname{sign}(ab)))\operatorname{sign}(ab)\Big).$$

Since $\operatorname{sign}(ab) \in \{-1, 1\}$, we have

$$\arccos(1) = 0, \qquad \arccos(-1) = \pi,$$

and therefore $\sin(\arccos(\operatorname{sign}(ab))) = 0$ in both cases. Hence

$$g(a,b) = \frac{|a||b|}{2\pi}(\pi - \arccos(\operatorname{sign}(ab)))\operatorname{sign}(ab).$$

If $ab > 0$, then $\operatorname{sign}(ab) = 1$ and

$$g(a,b) = \frac{|a||b|}{2\pi}(\pi - 0) = \frac{|a||b|}{2}.$$

If $ab < 0$, then $\operatorname{sign}(ab) = -1$ and

$$g(a,b) = \frac{|a||b|}{2\pi}(\pi - \pi)(-1) = 0.$$

Since $|a||b| = ab$ whenever $ab > 0$, this is exactly

$$g(a,b) = \frac{1}{2}\max(ab, 0).$$

Now if $M = \lambda_1 v_1 v_1'$, then

$$x'My = \lambda_1(v_1'x)(v_1'y), \qquad x'Mx = \lambda_1(v_1'x)^2, \qquad y'My = \lambda_1(v_1'y)^2.$$

Substituting these expressions into the definition of K_M gives

$$K_M(x,y) = \lambda_1\, g(v_1'x, v_1'y),$$

which proves the result. □

As a consequence, replacing the learned metric by its rank-one approximation $M^{(1)} =$

$\lambda_1 v_1 v_1'$ yields

$$
\begin{aligned}
f_T(x; T_M, z_M, T, z, \kappa = 1) &= \sum_{t=1}^{T} K_{M^{(1)}}(x, X_{t-1})\, R_t\, \xi_t \\
&= a(T_M, z_M, T, z)\ \max(v_1' x, 0) \\
&\quad + b(T_M, z_M, T, z)\ \max(-v_1' x, 0),
\end{aligned}
\tag{74}
$$

for suitable time-varying coefficients $a(\cdot)$ and $b(\cdot)$ determined by the historical sample. Thus a single rank-one RPM component is pseudo-linear in the sense used in the empirical section: it depends on characteristics only through the scalar index $v_1' x$, although the dependence on that scalar remains nonlinear.

E Numerical Implementation

This section describes the numerical implementation used to evaluate the kernel-based objects entering the estimator. The empirical application combines large cross sections with rolling estimation windows, so a naive implementation based on all pairwise stock-level kernel evaluations is computationally infeasible. The implementation therefore follows two principles throughout. First, it computes only the aggregated quantities required by the estimator rather than the full observation-level kernel matrices. Second, it organizes the required kernel evaluations into memory-efficient blocks that are well suited to GPU execution.

E.1 Portfolio Kernel Computation

For dates t and s, the portfolio kernel is

$$
\mathbb{K}_{t,s} = \frac{1}{\sqrt{N_t N_s}} \sum_{i=1}^{N_t} \sum_{j=1}^{N_s} R_{i,t+1}\, K(X_{i,t}, X_{j,s})\ R_{j,s+1}.
$$

A naive implementation would attempt to form the full observation-level kernel matrix over all stock–month observations in a rolling window. In the present application, each month contains roughly 3000 stocks, prediction windows extend up to 360 months, and update windows extend up to 240 months. A naive observation-level kernel matrix would therefore involve on the order of one million stock–month observations, making both storage and computation infeasible.

The key simplification is that the estimator requires only the aggregated quantity $\mathbb{K}_{t,s}$, not the underlying matrix of all pairwise stock-level kernel values. The computation is therefore organized so that kernel evaluations are generated only when needed and are immediately contracted into lower-dimensional sufficient statistics.

Fix a date t. The objective is to compute

$$\mathbb{K}_{t,s} = \frac{1}{\sqrt{N_t N_s}} \sum_{i=1}^{N_t} \sum_{j=1}^{N_s} R_{i,t+1}\, K(X_{i,t}, X_{j,s})\, R_{j,s+1} \qquad \text{for all } s = 1, \ldots, T.$$

To do so, define the intermediate quantity

$$h_{t,j} = \sum_{i=1}^{N_t} R_{i,t+1}\, K\big(X_{i,t}, X_{j,\tau(j)}\big)\,,$$

where $\tau(j)$ denotes the time index of observation j within the rolling window. Once $h_{t,j}$ has been computed for all observations j, the portfolio-kernel row can be written as

$$\mathbb{K}_{t,s} = \frac{1}{\sqrt{N_t N_s}} \sum_{j:\, \tau(j)=s} h_{t,j}\, R_{j,s+1}.$$

This separates the computation into two stages: first contract over the left cross section, and then aggregate over the right cross section by time index.

The implementation uses a two-sided blocking scheme. On the left side, the observations in cross section t are partitioned into blocks of size

$$B_i^{\text{PTK}} = 4000.$$

On the right side, all observations in the rolling window are partitioned into blocks of size

$$B_j^{\text{PTK}} = 160000.$$

For each pair consisting of one left block and one right block, the algorithm evaluates the corresponding block kernel matrix

$$K_{ij} = K(X_{i,t}, X_{j,\tau(j)}),$$

which has dimension $B_i^{\text{PTK}} \times B_j^{\text{PTK}}$. This block is then immediately contracted with the left-side returns, so only the block contribution to the aggregated object is retained. The resulting vector is then multiplied by the right-side returns and accumulated by time index to update the partial values of $\mathbb{K}_{t,s}$.

The full portfolio kernel is obtained by repeating this contraction-and-aggregation procedure over all right blocks for a given left block, then over all left blocks for the fixed date t, and finally over all dates $t = 1, \ldots, T$. The largest matrix ever materialized is therefore a block-level kernel matrix of size

$$B_i^{\mathrm{PTK}} \times B_j^{\mathrm{PTK}},$$

rather than the full observation-level kernel matrix. This reduces peak memory usage dramatically and makes the computation efficient on the GPU.

E.2 Normalization and Numerical Stabilization

The implementation uses two normalization steps together with a small numerical safeguard.

First, whenever denominators or square roots may become numerically small, we add a constant

$$\varepsilon = 10^{-8}.$$

This prevents undefined operations and reduces sensitivity to near-zero scaling terms.

Second, in the empirical specification reported in the paper, each monthly update vector is normalized by its Euclidean norm,

$$\tilde{g}_s = \frac{g_s}{\|g_s\|_2},$$

before aggregation across months. The rolling learned metric is then formed as the average of the outer products $\tilde{g}_s \tilde{g}_s'$.

Finally, after aggregation we apply trace normalization,

$$M_{1,t} \leftarrow \frac{M_{1,t}}{\mathrm{tr}(M_{1,t})}.$$

Under exact arithmetic this final step is redundant once each $\tilde{g}_s \tilde{g}_s'$ has trace one, but we retain it as a numerical normalization to keep the learned metric on a stable scale across rolling windows.

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